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AIM: Performing Data Analysis using Calculations
in Tableau

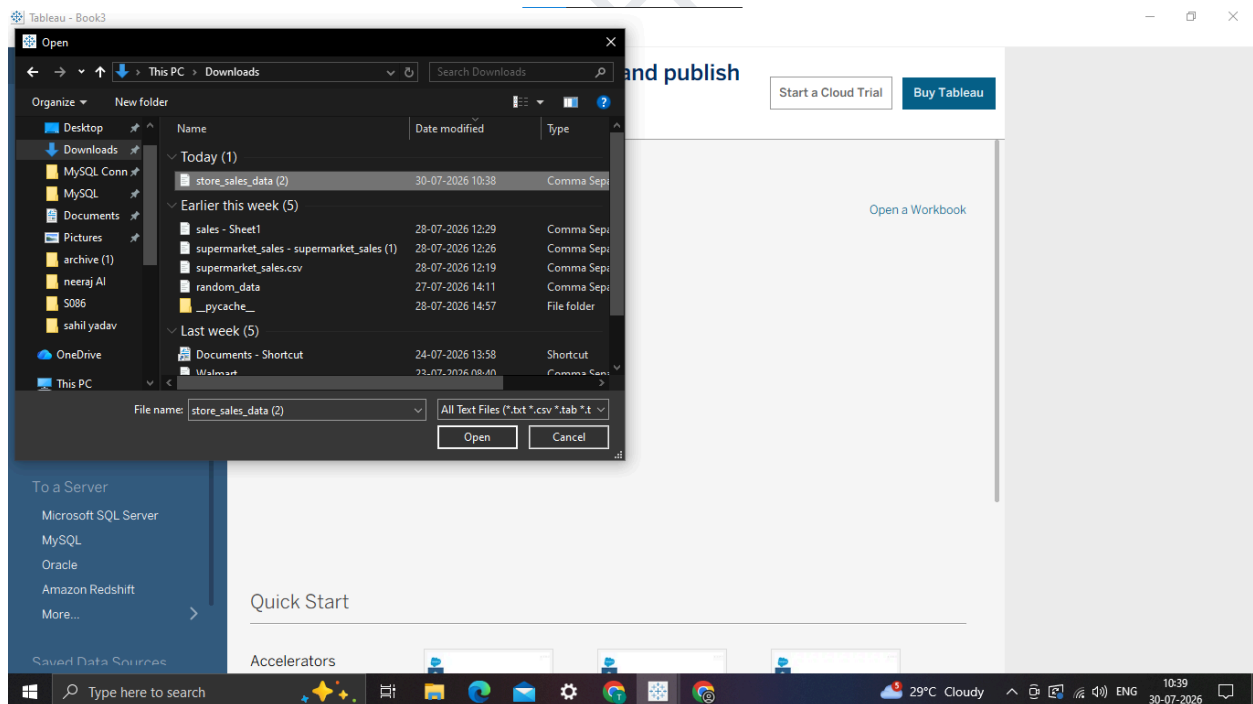
- a) Create basic calculated fields using formulas
- b) Perform row-level calculations
- c) Perform aggregate-level calculations
- d) Implement FIXED Level of Detail (LOD) calculations
- e) Use INCLUDE and EXCLUDE LOD expressions
- f) Create and use parameters for dynamic analysis
- g) Apply calculations to fix data issues and enhance insights

Dataset: Indian Store Data

A: Creating Basic Calculated Fields Using Formulas

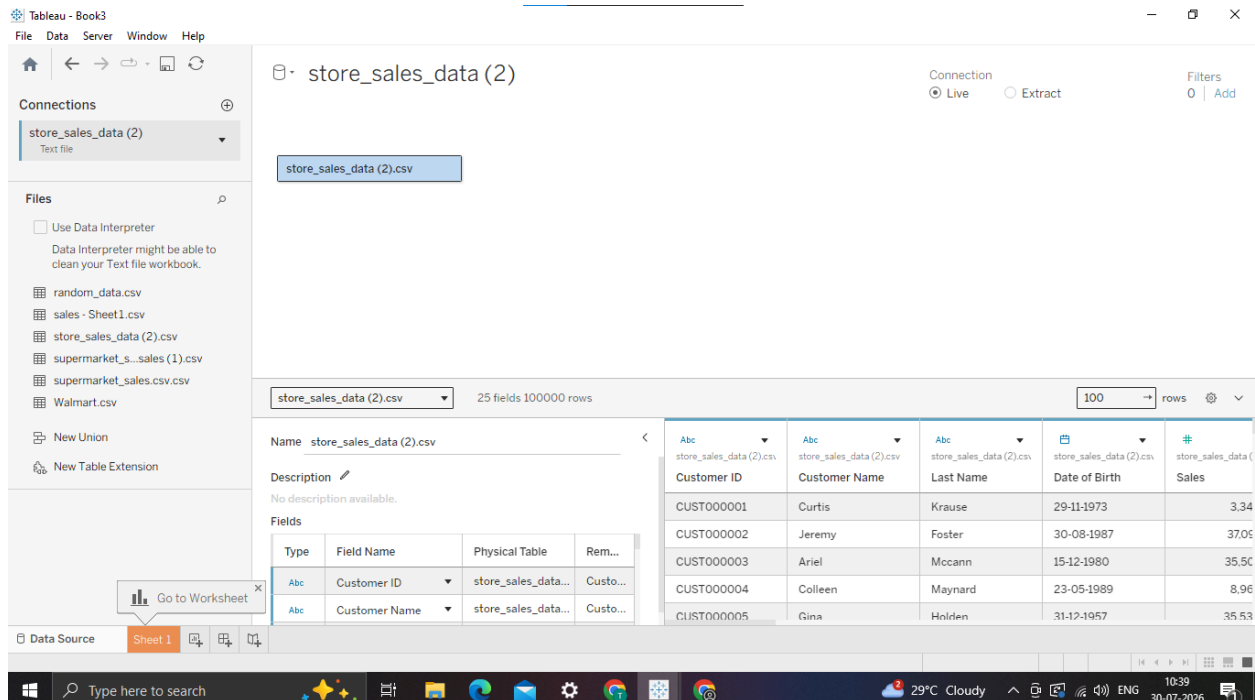
1. Connecting to the Dataset

1. Launch Tableau. Under the Connect panel, click on Text File, select your store_sales_data.csv, and click Open.



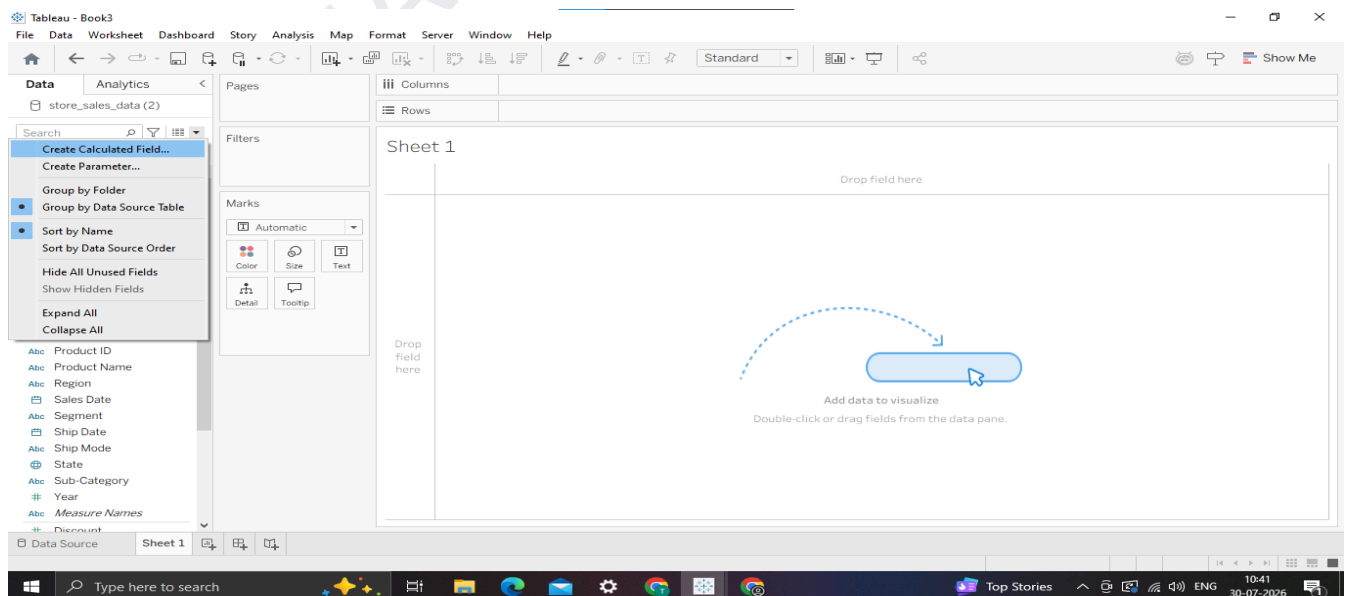
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2. Click on Sheet 1 at the bottom-left corner to enter your workspace.



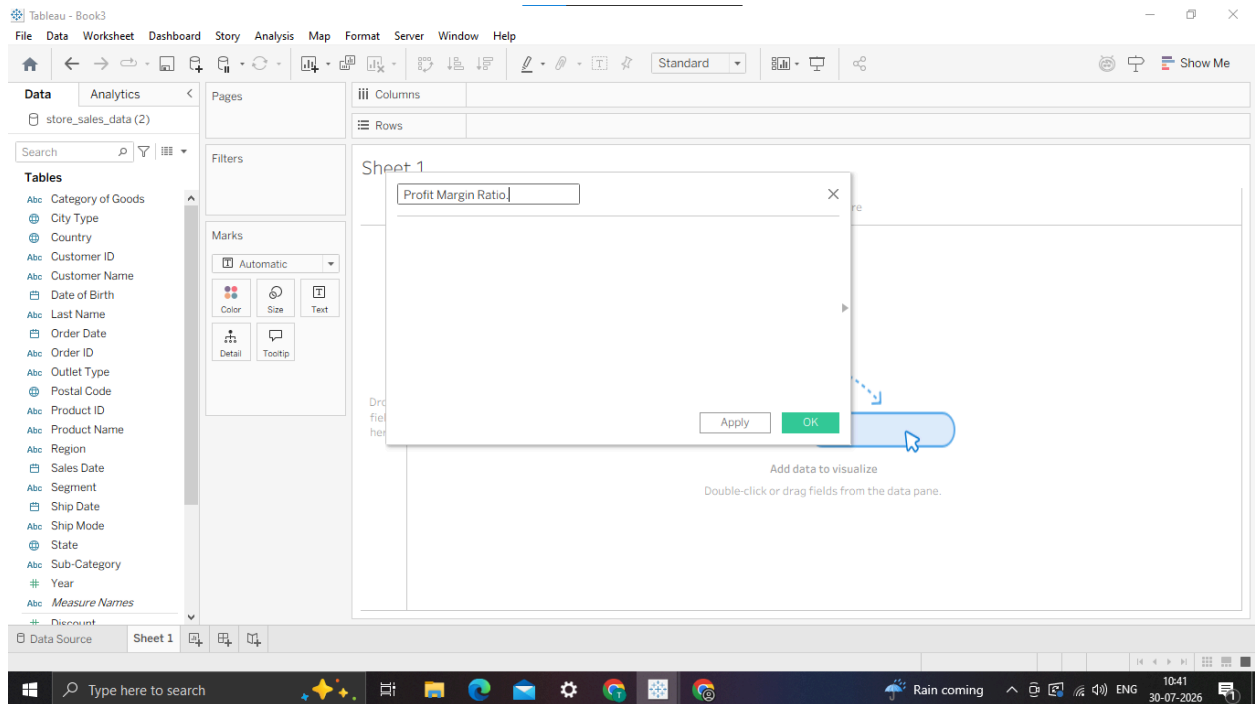
2. Building a Basic Custom Calculation

1. Look at your left-hand Data Pane. Right-click anywhere in the blank space of the pane (or click the small downward triangle next to the search bar) and select Create Calculated Field...

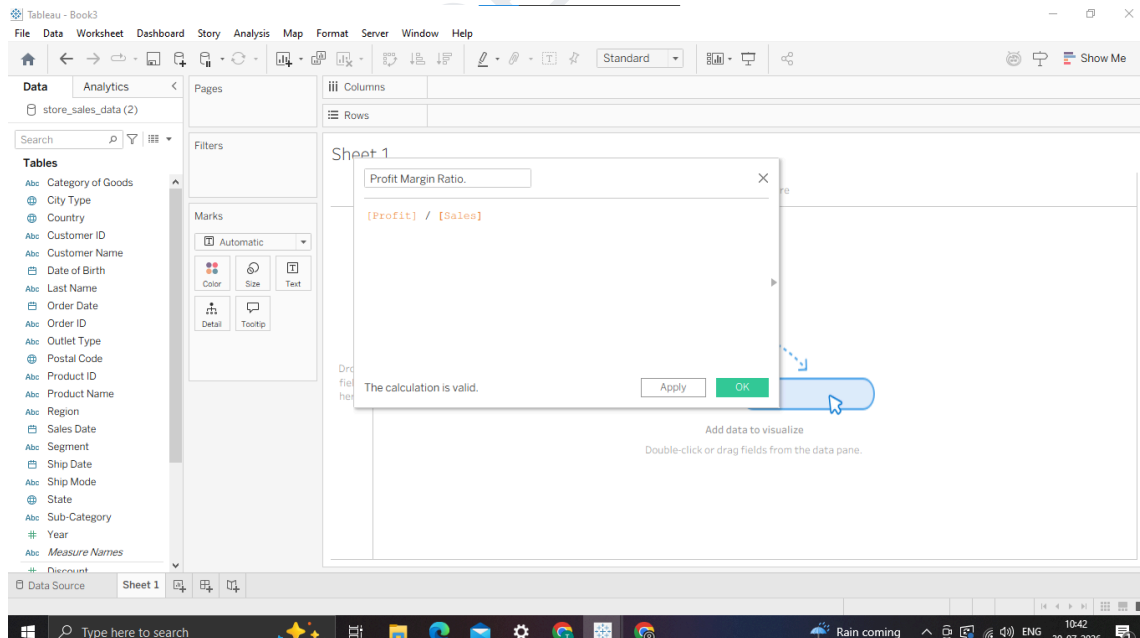


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2. In the calculation editor window that pops up, clear any default text and name the field: Profit Margin Ratio.

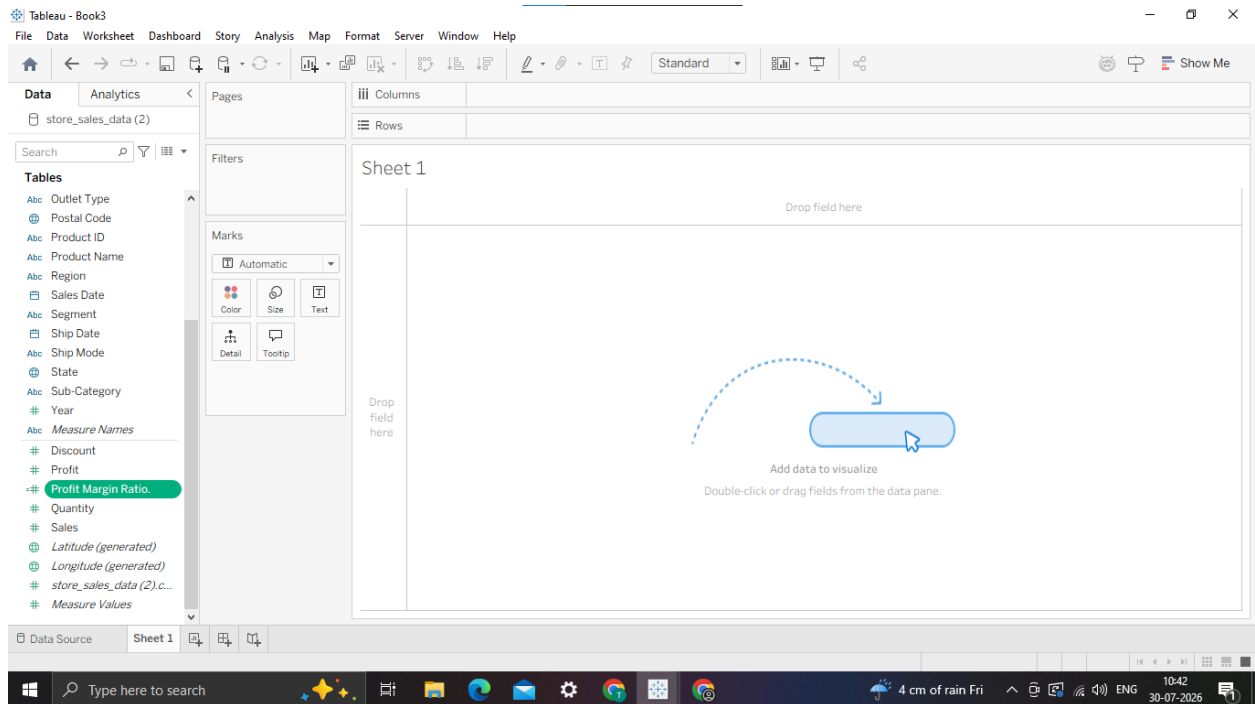


3. In the main workspace box, enter the following basic mathematical formula: [Profit] / [Sales]

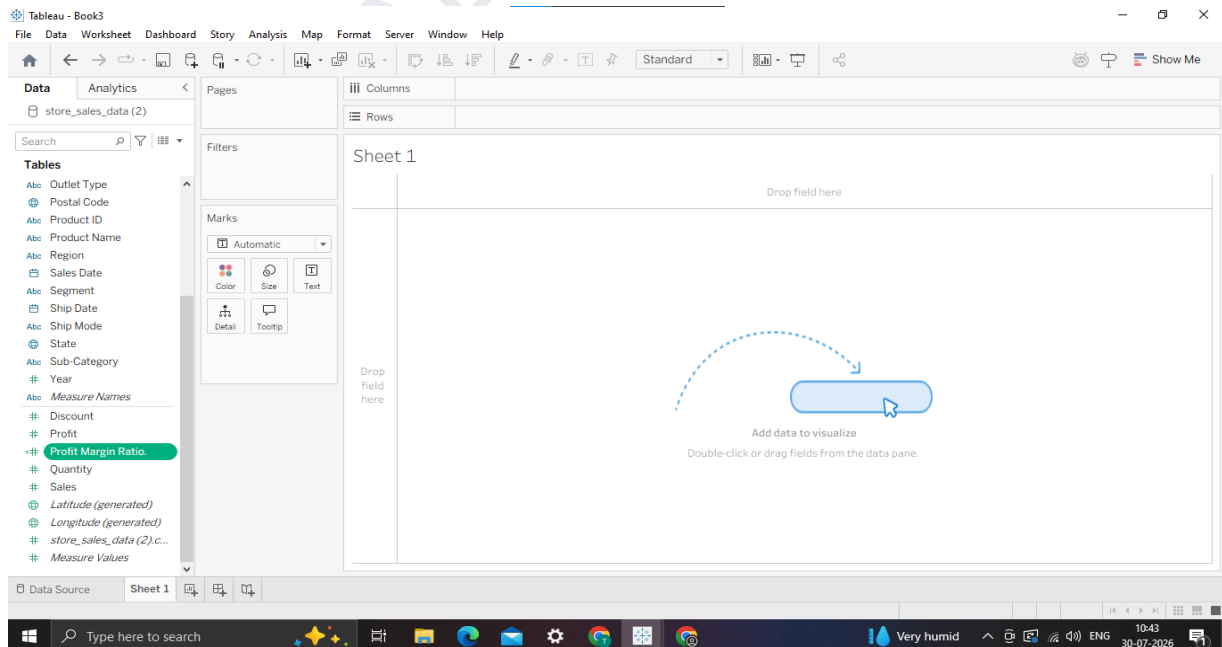


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4. Look at the bottom-left corner of the editor box. It should say "The calculation is valid."
Click OK.



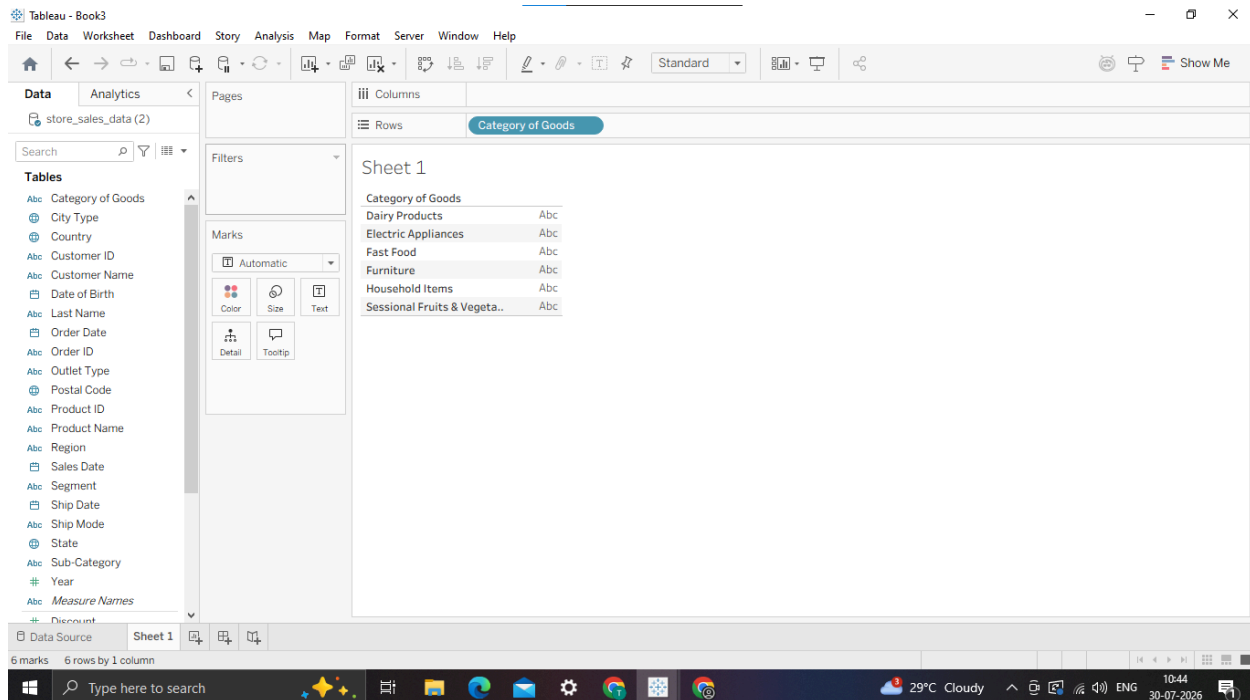
5. Notice that your new custom field Profit Margin Ratio now appears in your Data Pane list with a small equal sign (=) next to its data type icon, showing it is a calculated field.



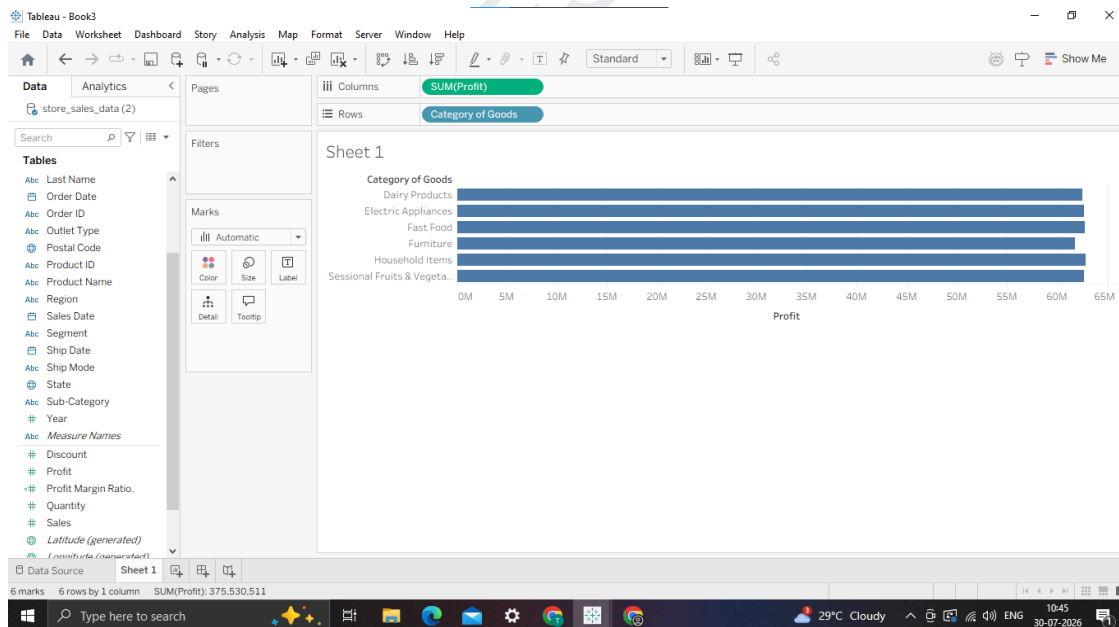
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2. Putting the Field to Use

1. Drag the Category of Goods dimension from your Data Pane and drop it onto the Rows shelf.

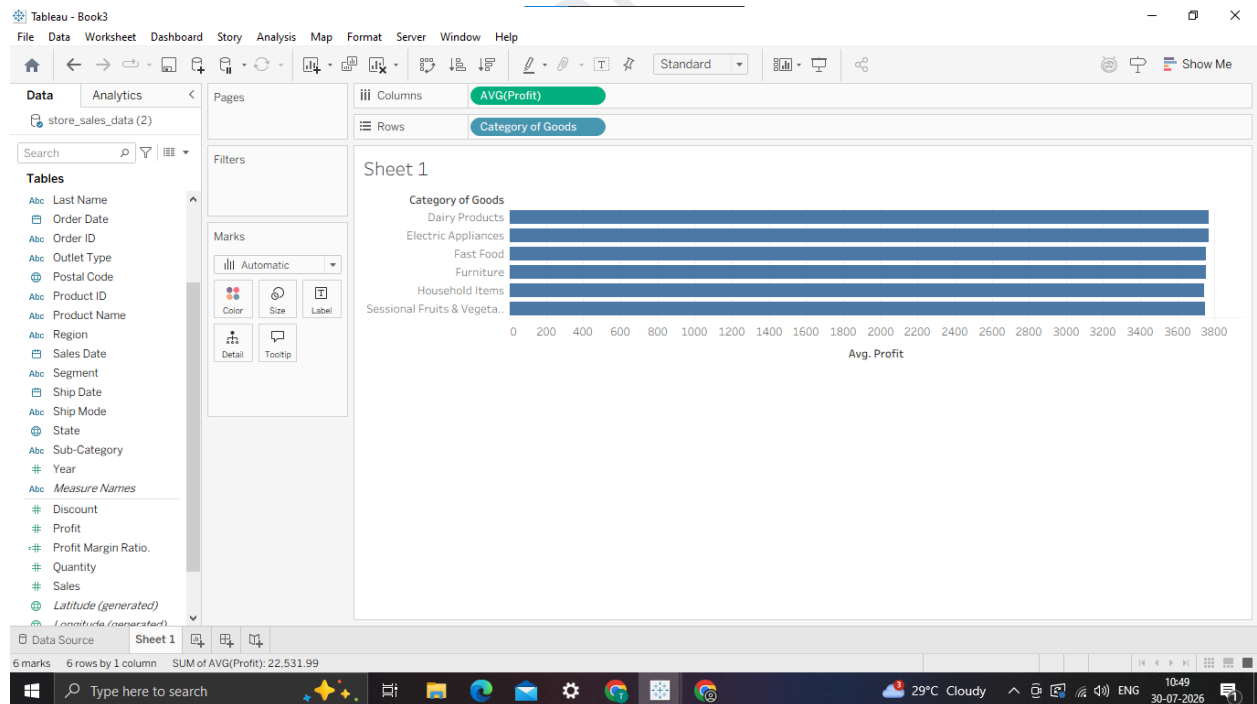
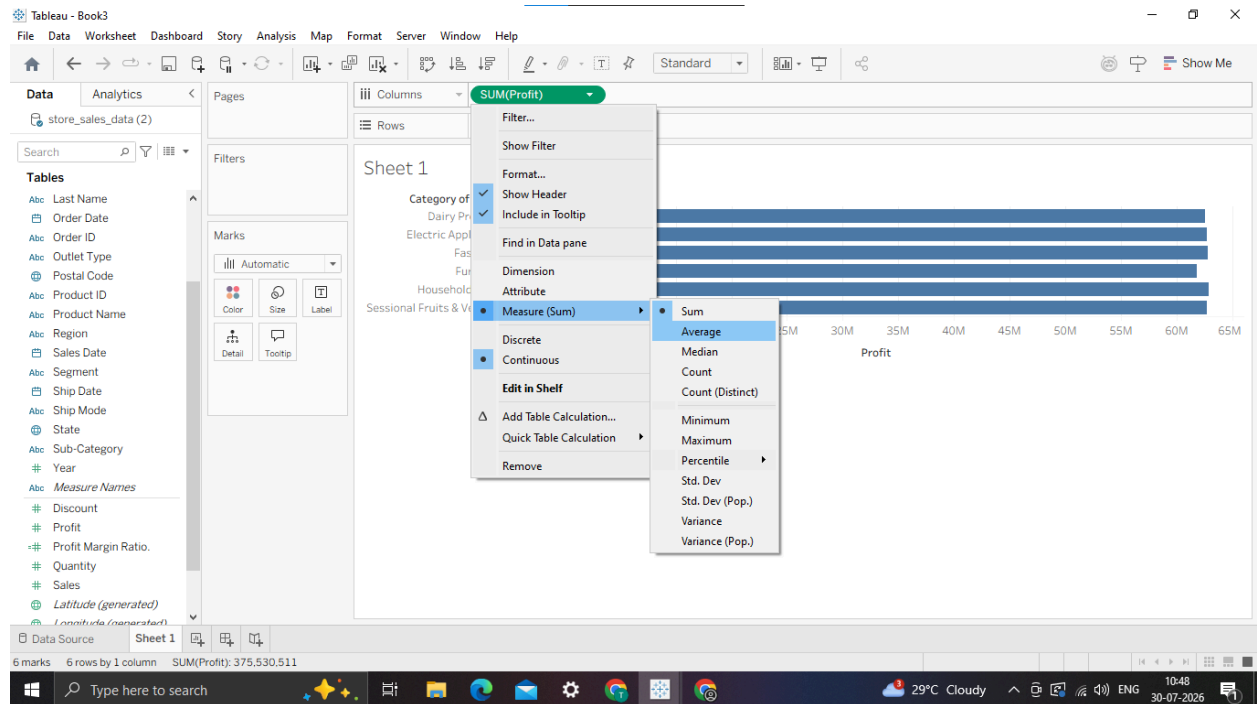


2. Drag your newly created Profit Margin Ratio calculated field and drop it onto the Columns shelf.



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3. Right-click the Profit Margin Ratio pill now sitting on the Columns shelf, hover over Measure (Sum), and change it to Average to see the true margin distribution across product types.

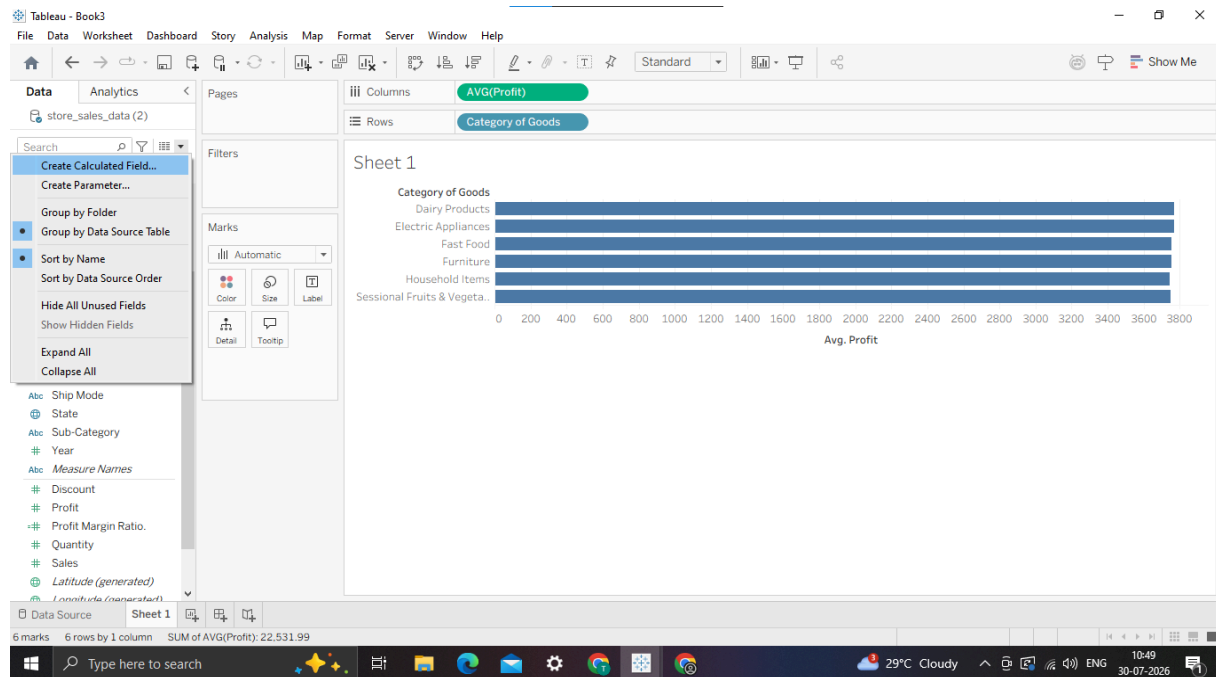


B: Performing Row-Level Calculations

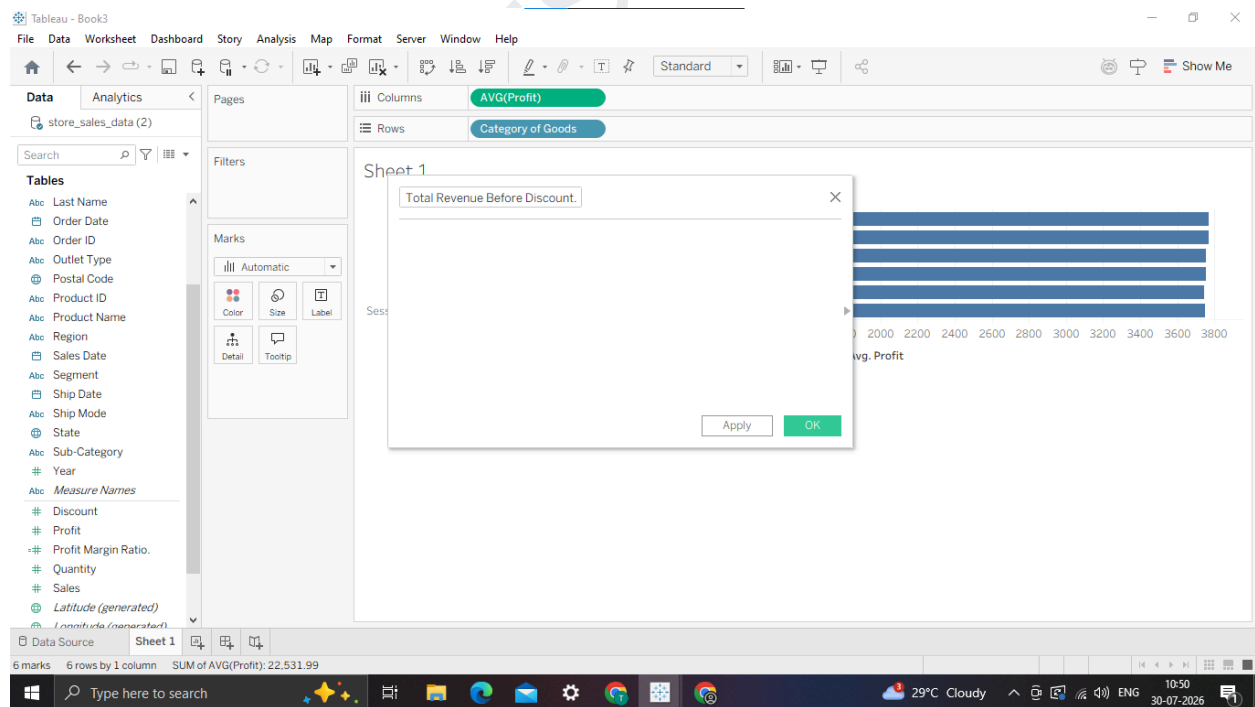
1. Creating a Row-Level Calculated Field

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1. Right-click anywhere in the blank white space of your Data Pane and choose Create Calculated Field...

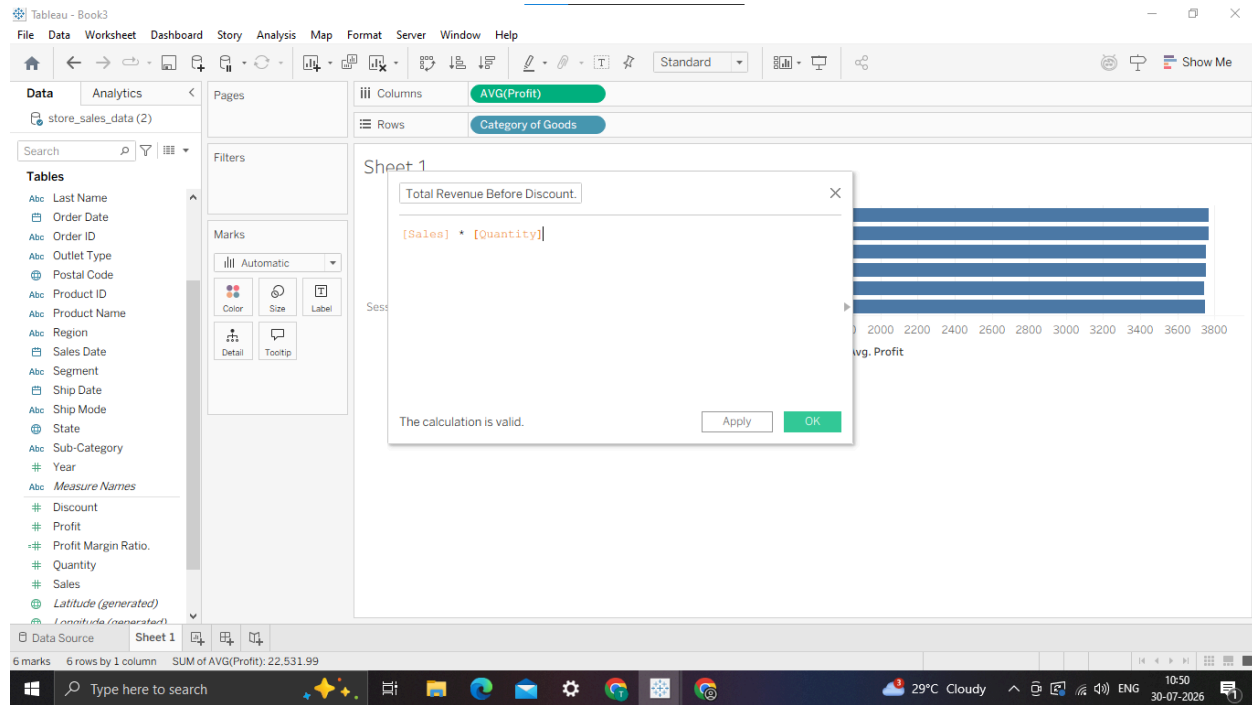


2. Name the field: Total Revenue Before Discount.

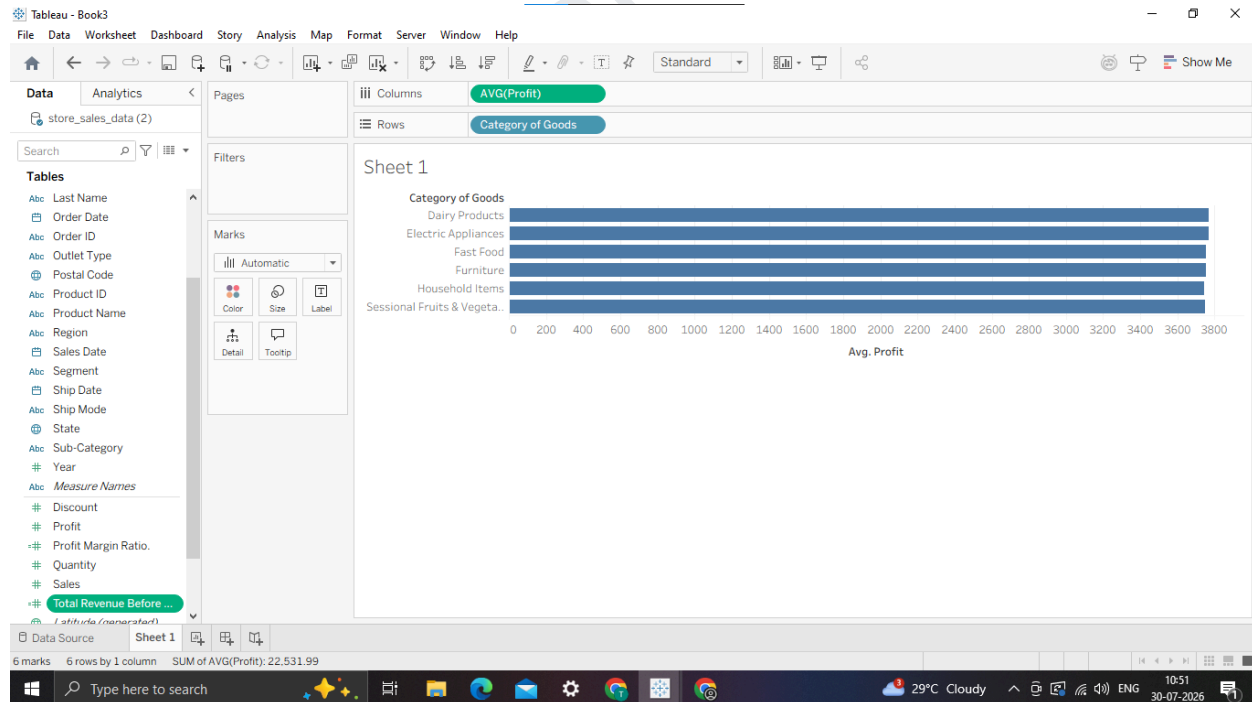


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3. In the calculation formula canvas area, input the following expression:



4. Verify the message at the bottom reads "The calculation is valid" and click OK.

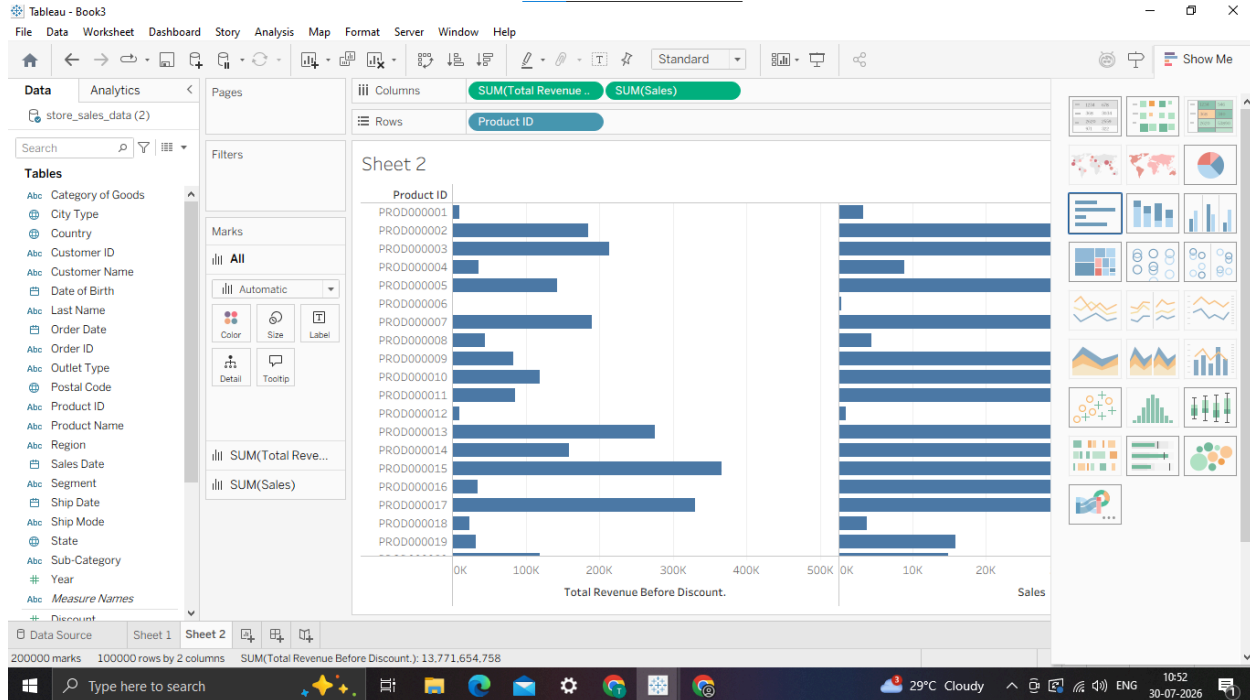


2. Putting the Field to Use

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1. Drag the Product Name dimension from the Data Pane and drop it onto the Rows shelf.
2. Drag your newly created Total Revenue Before Discount calculated field and drop it onto the Columns shelf.
3. Drag the raw Sales measure pill and drop it directly onto the Columns shelf right next to the previous calculated pill to visually compare the two numbers.

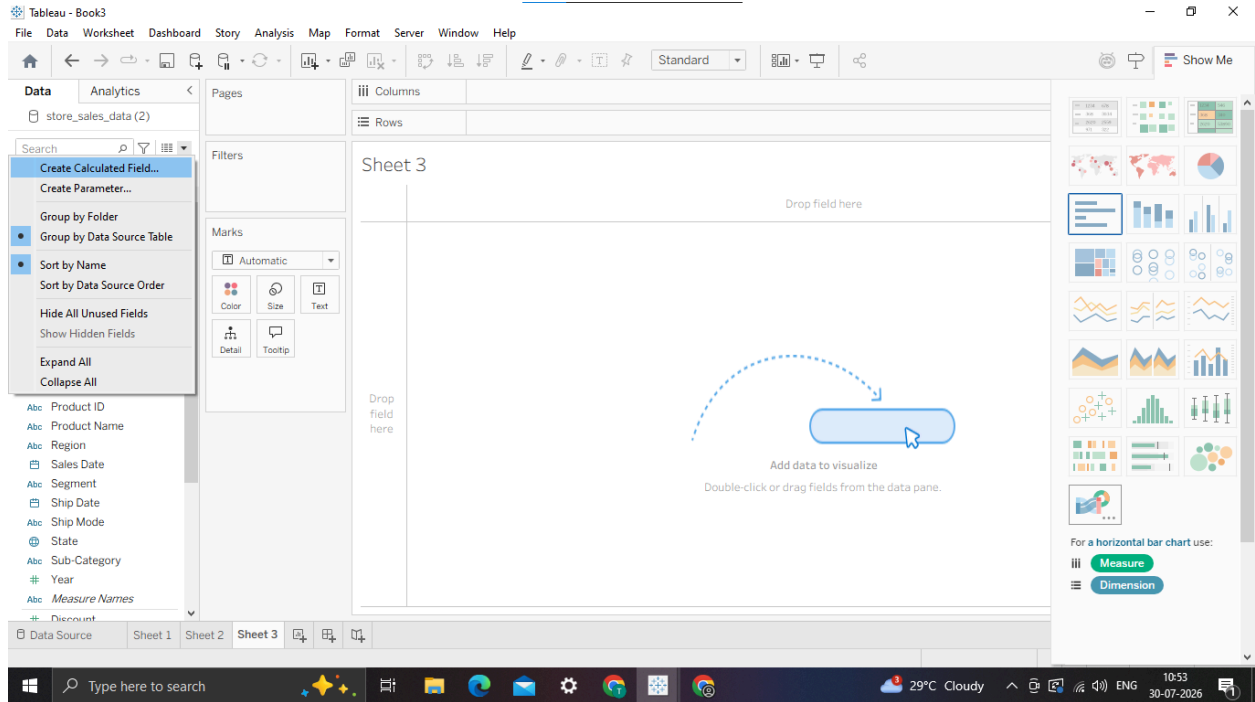


C: Performing Aggregate-Level Calculations

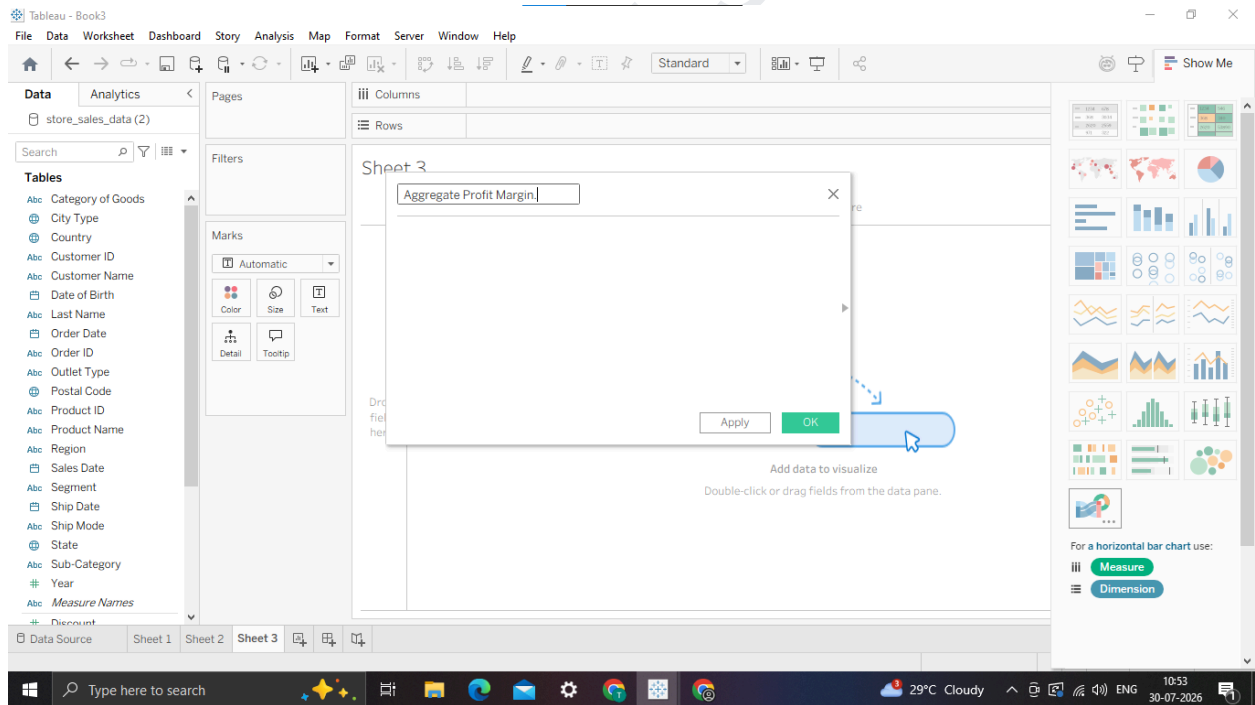
1. Creating an Aggregate Calculated Field

1. Right-click in the blank white space of your Data Pane or Click the Small Triangle Icon next to search bar and select Create Calculated Field...

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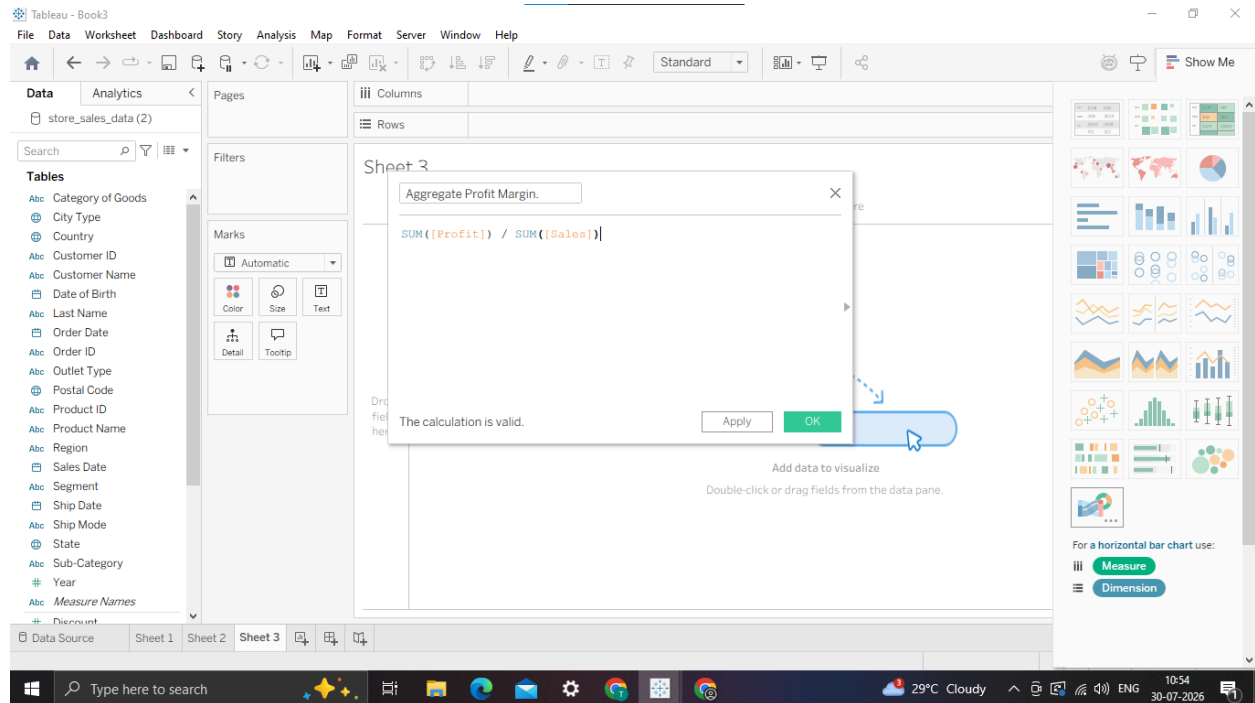


2. Name the field:

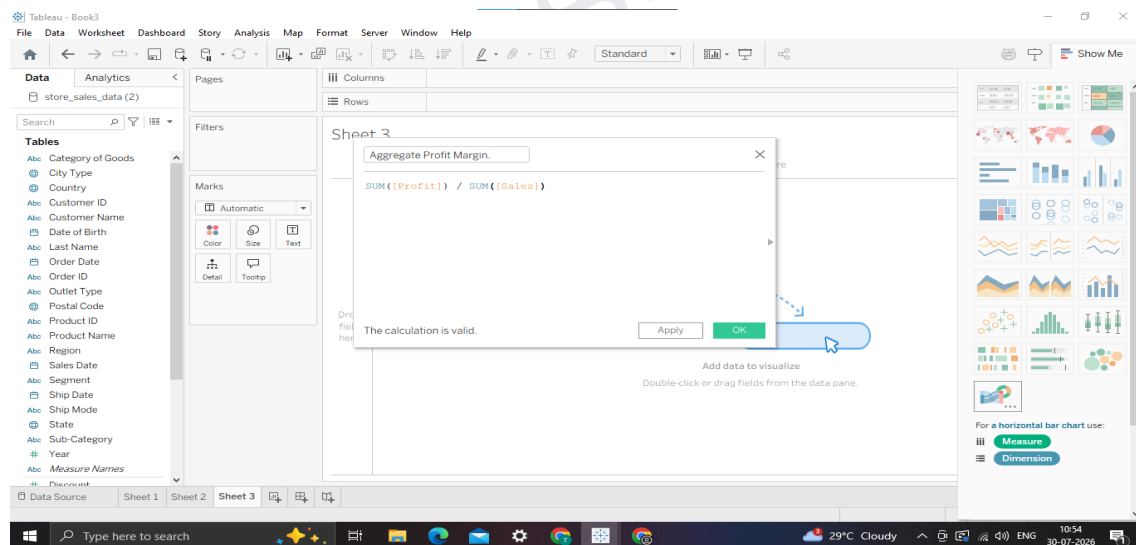


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3. In the calculation formula window, type the following expression:
 $\text{SUM}([\text{Profit}]) / \text{SUM}([\text{Sales}])$



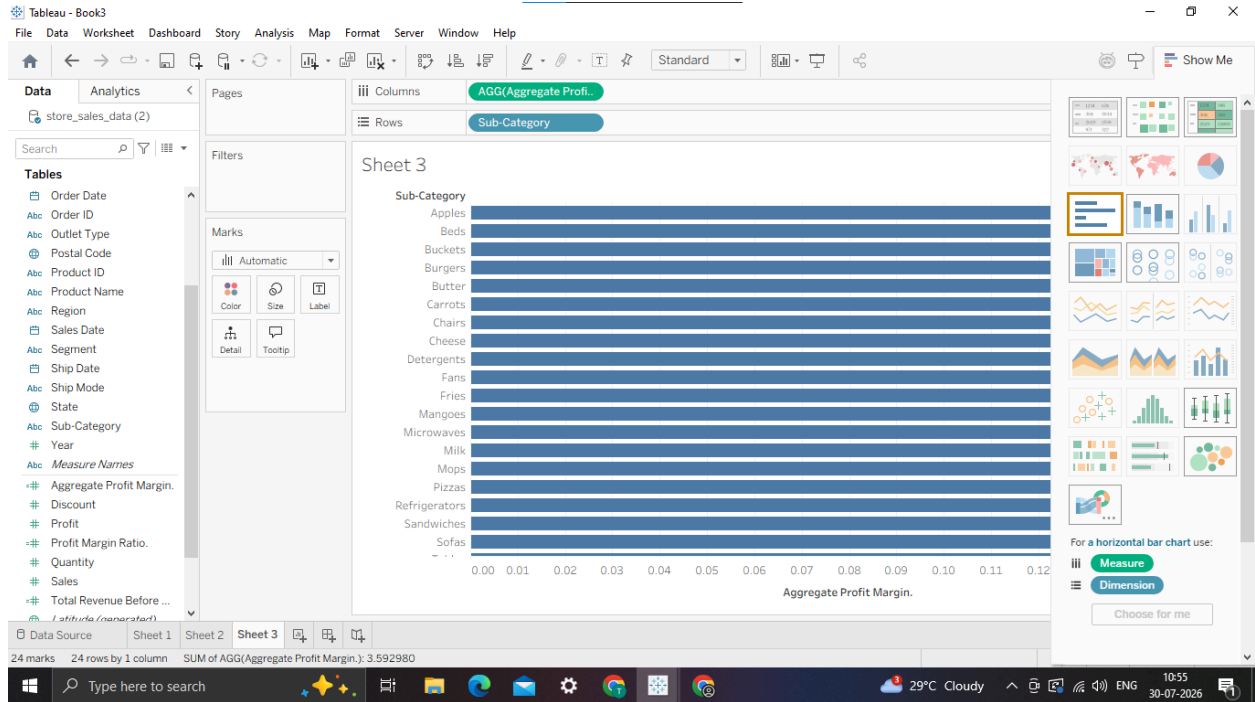
4. Look at the bottom left to ensure it reads "The calculation is valid" and click OK.



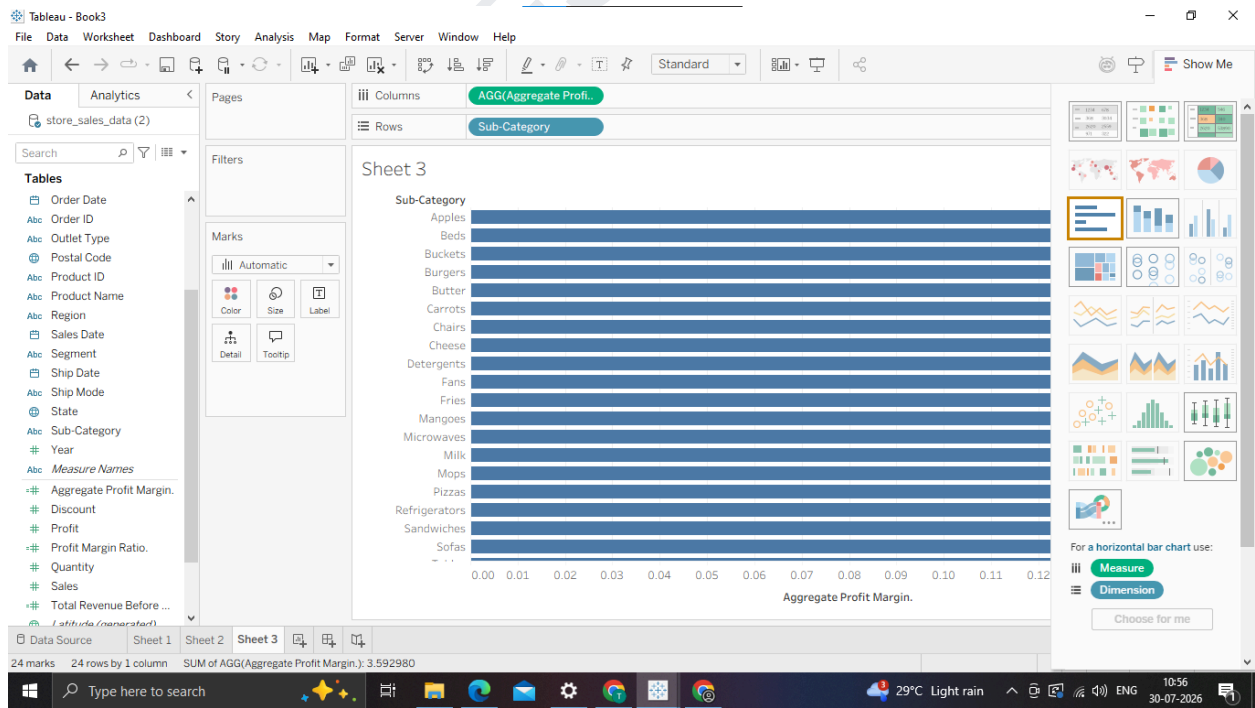
2. Putting the Field to Use

1. Drag the Sub-Category dimension from the Data Pane and drop it onto the Rows shelf.
2. Drag your newly created Aggregate Profit Margin calculated field and drop it onto the Columns shelf.

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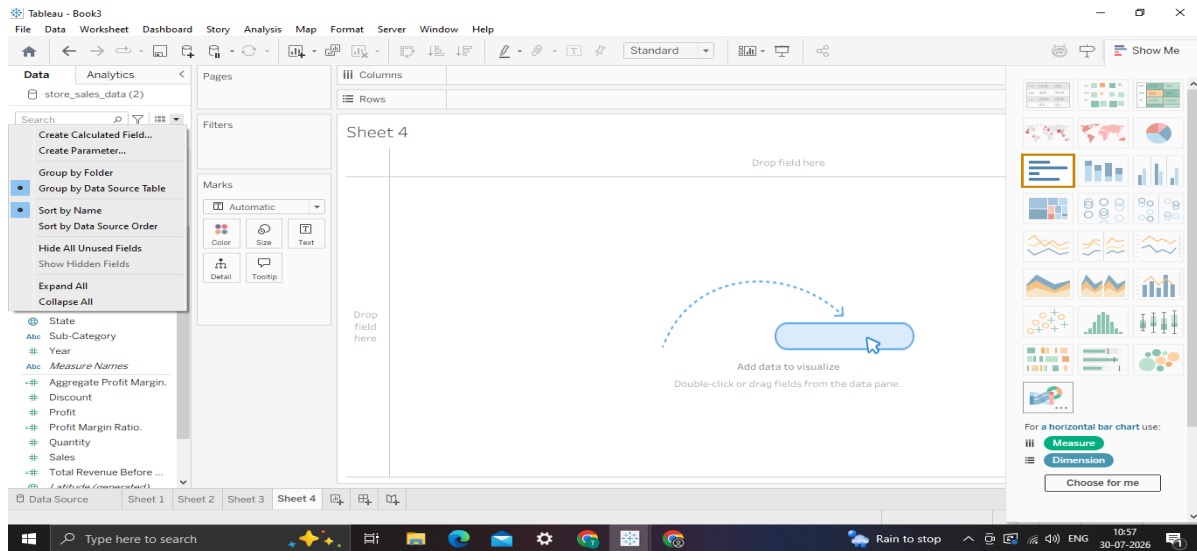
3. Observe closely: Look at the pill on the Columns shelf. It displays as AGG(Aggregate Profit Margin). Unlike standard fields, you cannot right-click this pill to change it to "Average" or "Count"—Tableau locks it as an aggregation because the math is already defined inside the formula.



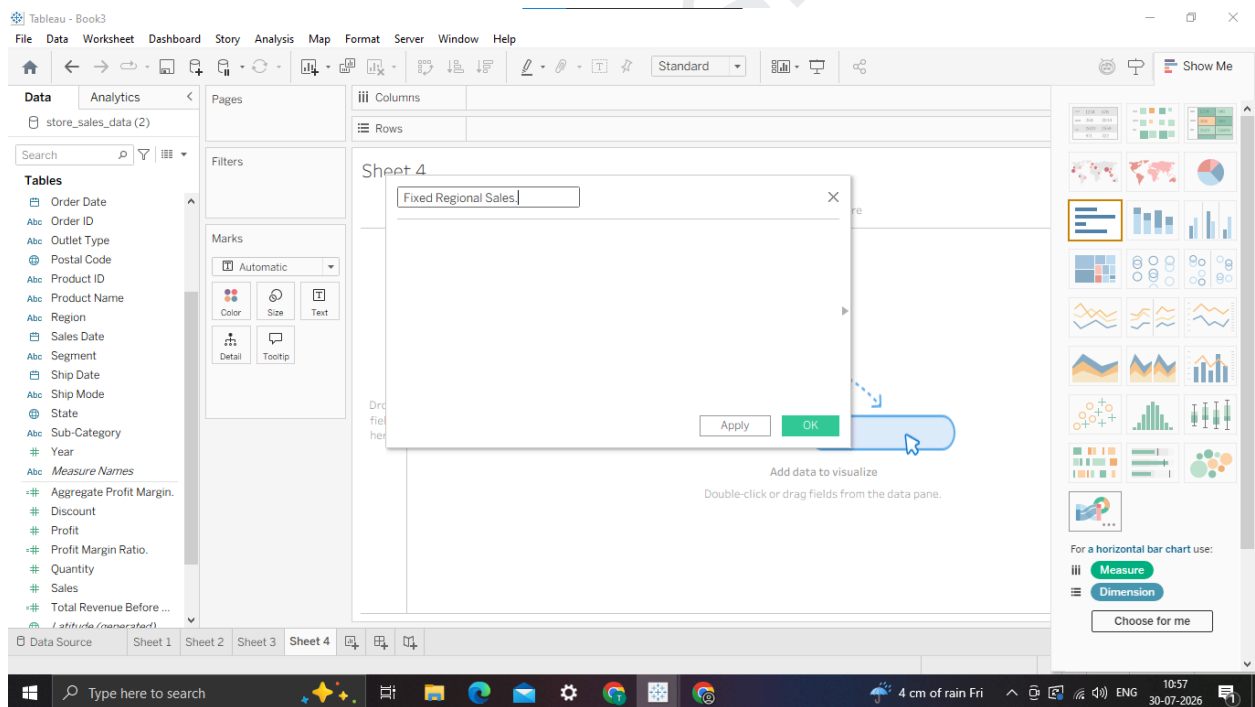
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1. Creating a FIXED LOD Calculated Field

1. Right-click in the blank white space of your Data Pane or Click the Small Triangle Icon next to search bar and select Create Calculated Field...



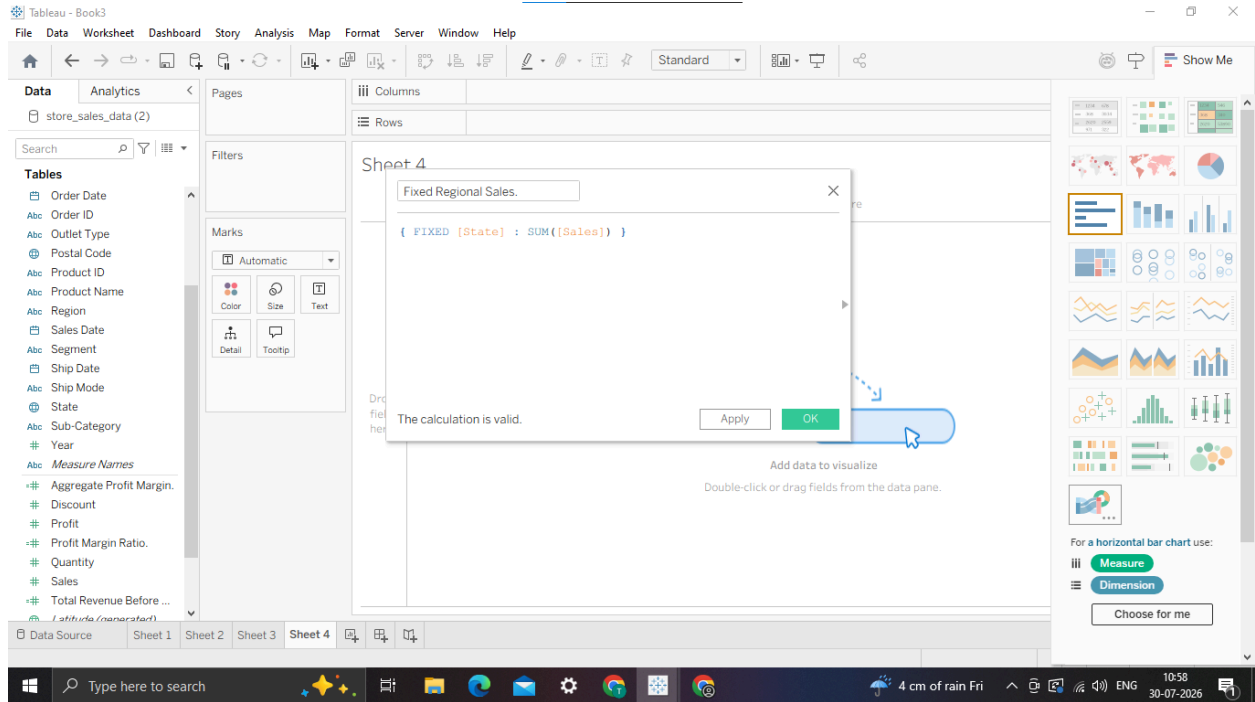
2. Name the field: Fixed Regional Sales.



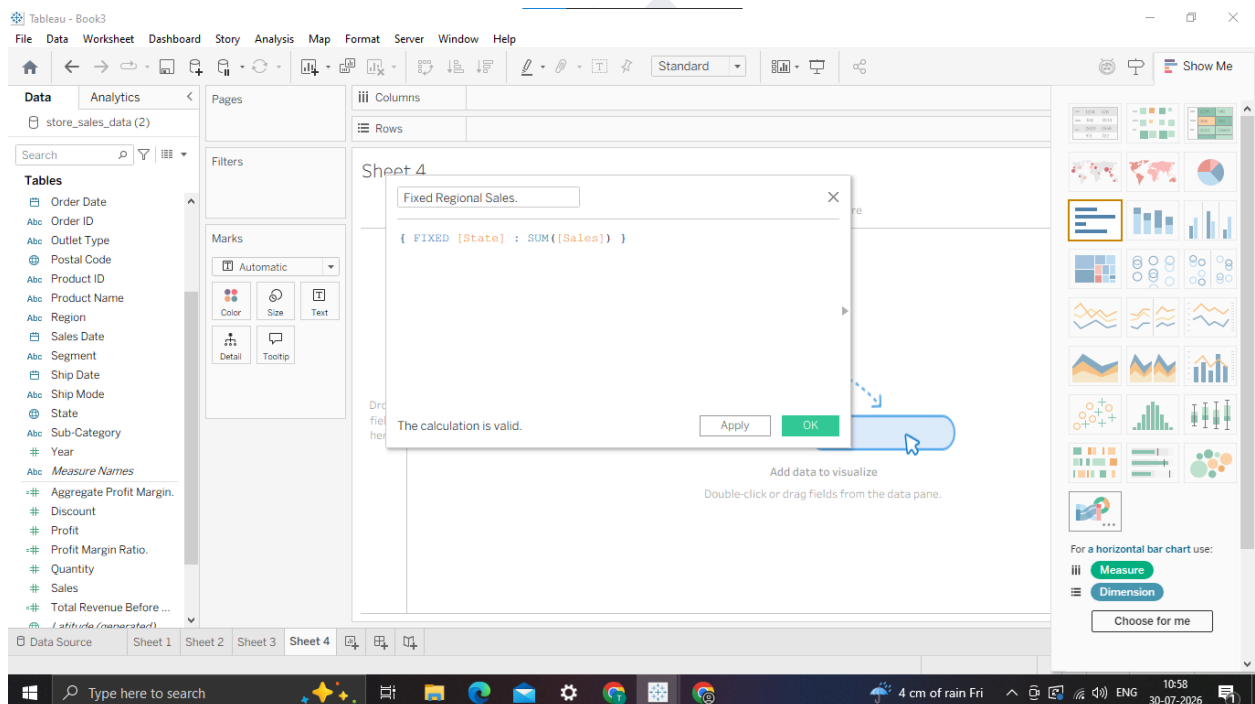
3. In the calculation formula window, type the following expression using curly braces

{:
{ FIXED [State] : SUM([Sales]) }

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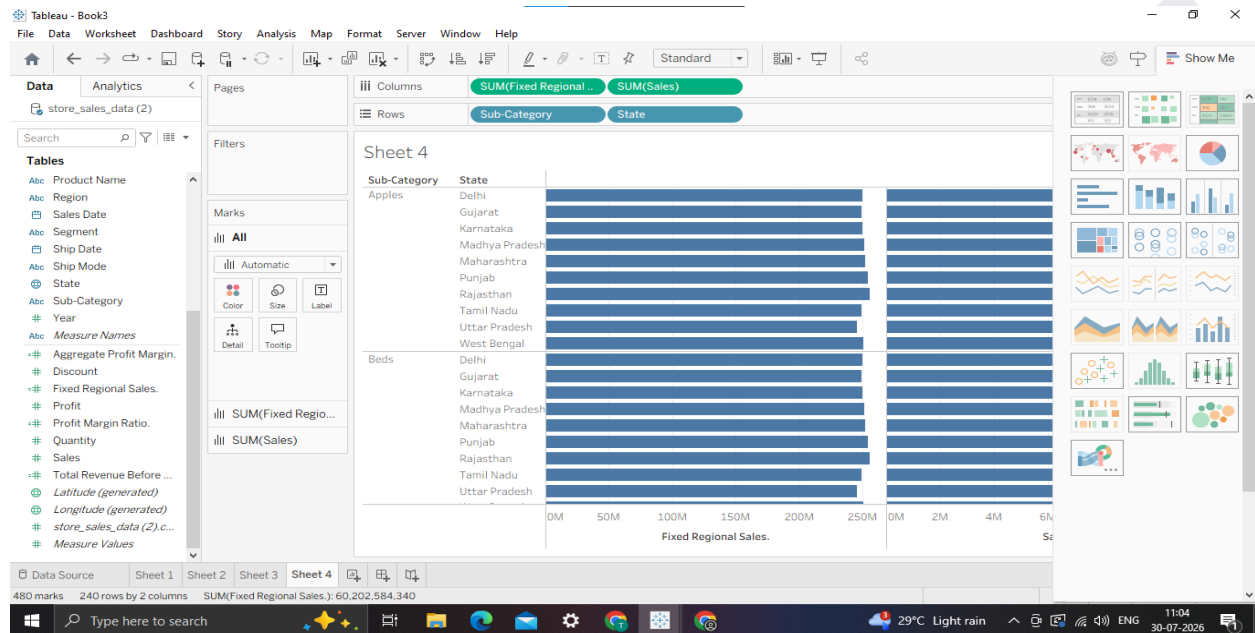
4. Look at the bottom left to ensure it reads "The calculation is valid" and click OK.



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2. Putting the Field to Use

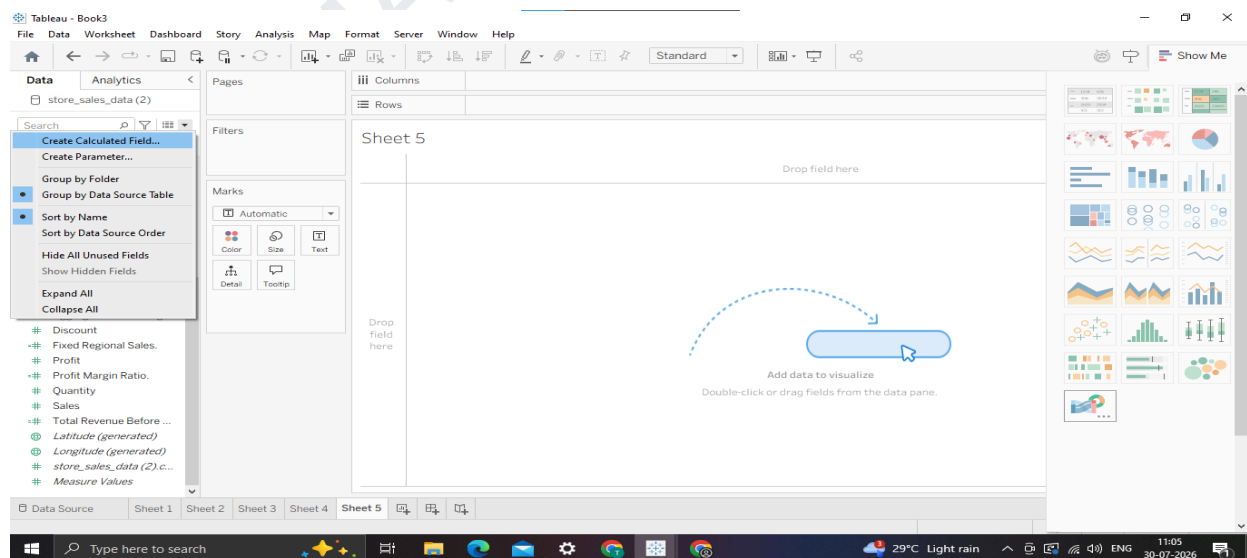
1. Drag the State dimension from the Data Pane and drop it onto the Rows shelf.
2. Drag the Sub-Category dimension and drop it onto the Rows shelf, right next to State.
3. Drag the raw Sales measure pill onto the Columns shelf. (This shows the sales split down by each sub-category within that state).
4. Drag your newly created Fixed Regional Sales calculated field and drop it onto the Columns shelf, right next to the raw Sales pill.



E: Using INCLUDE and EXCLUDE LOD Expressions

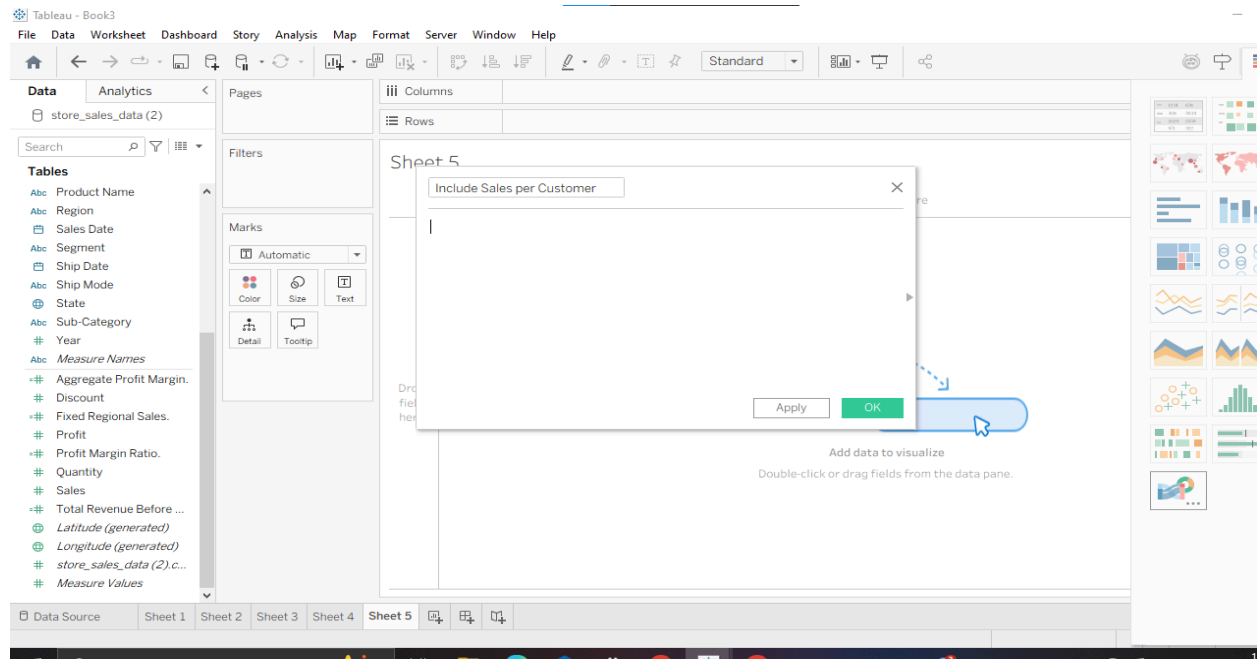
1. Creating an INCLUDE LOD Calculated Field

1. Right-click in the blank white space of your Data Pane and select Create Calculated Field...

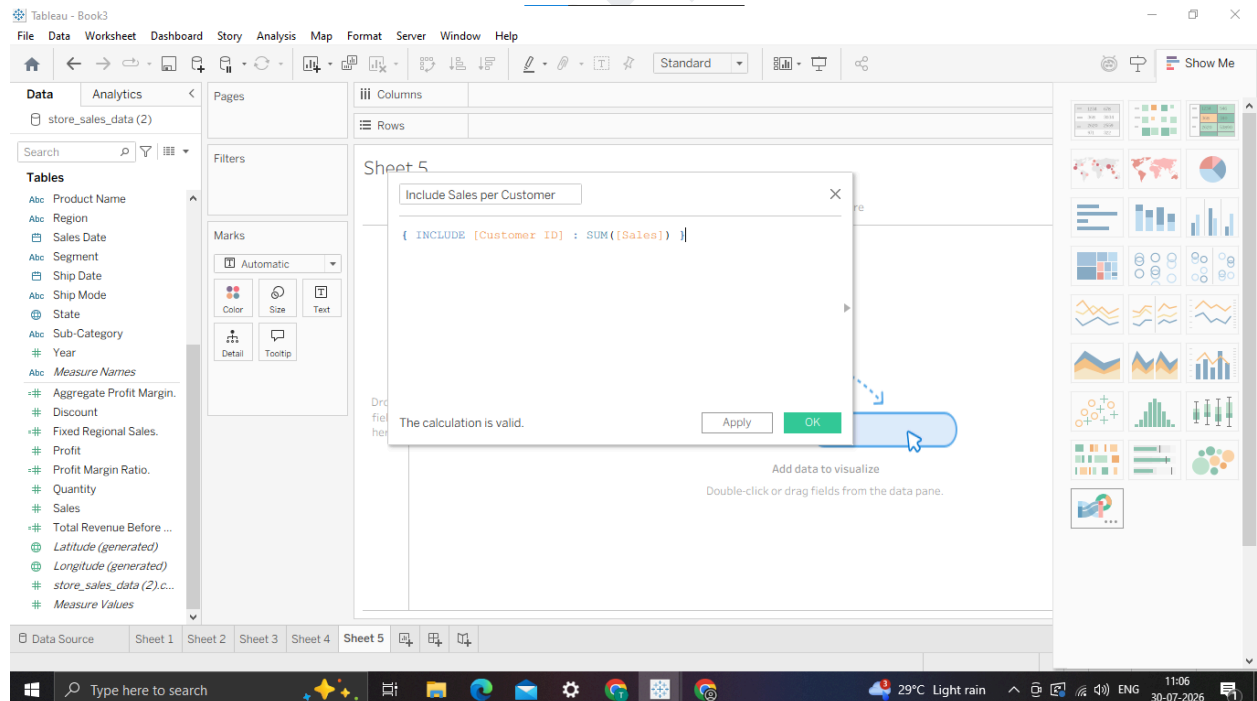


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2. Name the field: Include Sales per Customer.

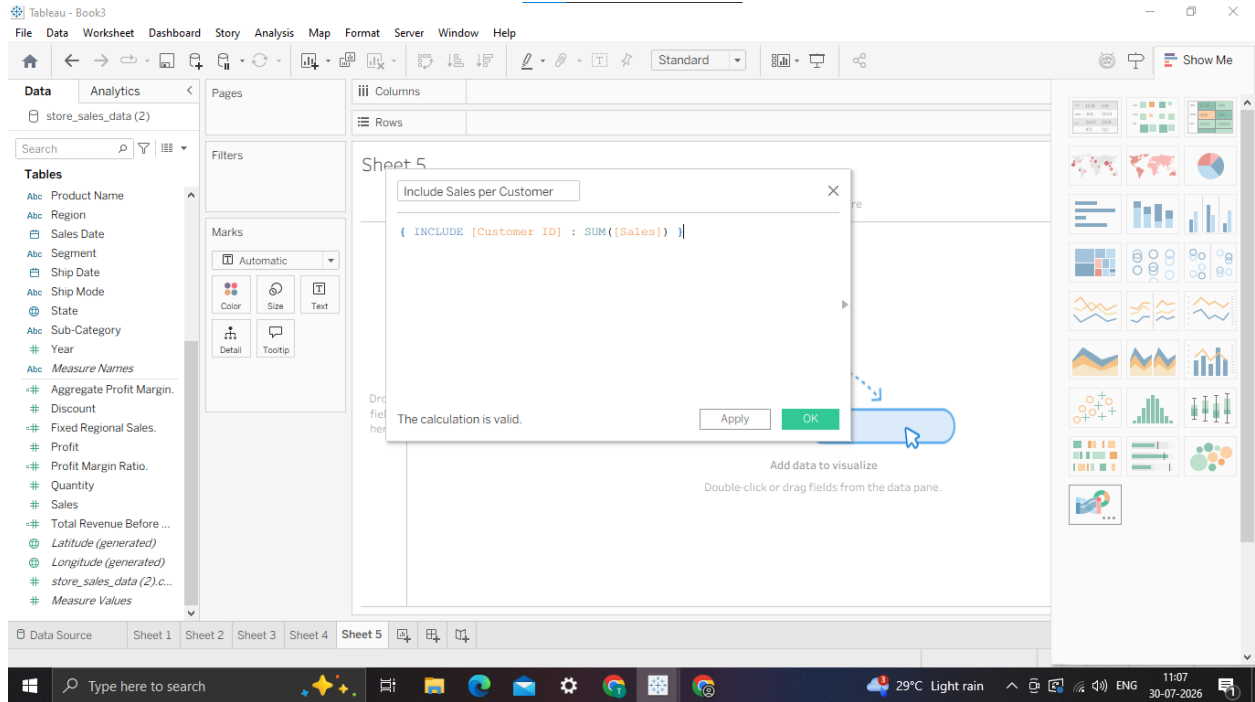


3. In the calculation formula canvas area, input the following expression: { INCLUDE [Customer ID] : SUM([Sales]) }



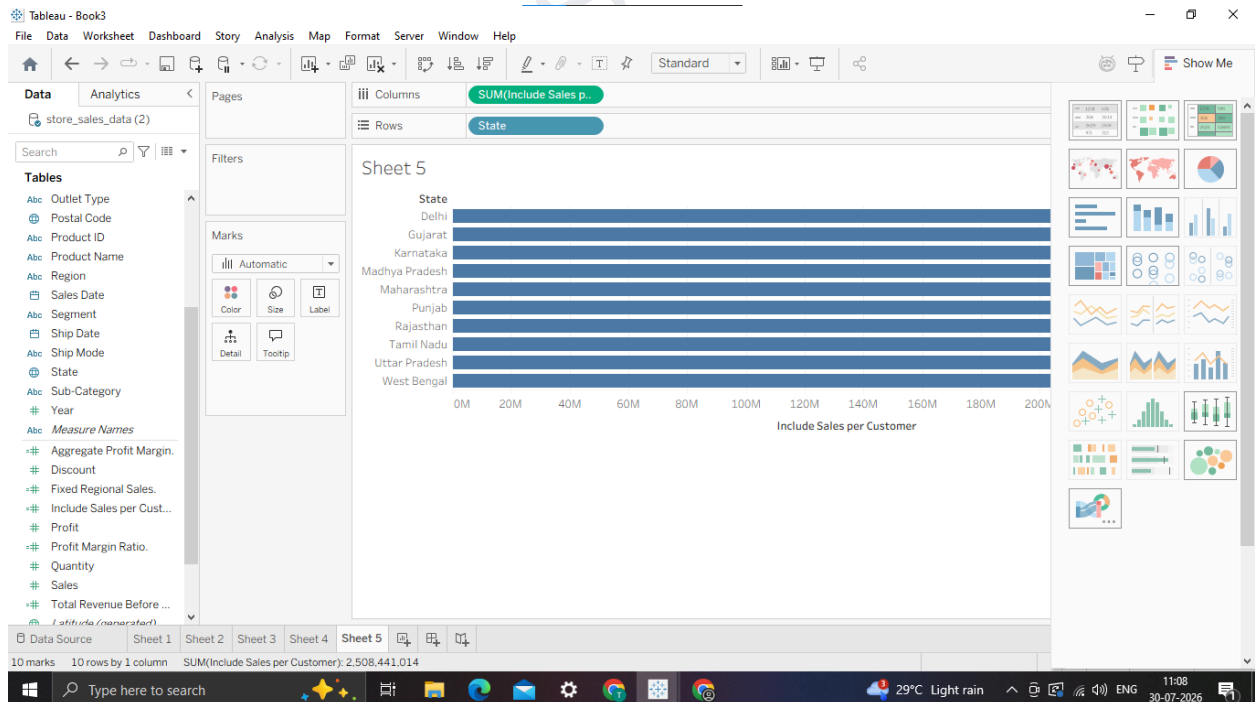
4. Click OK.

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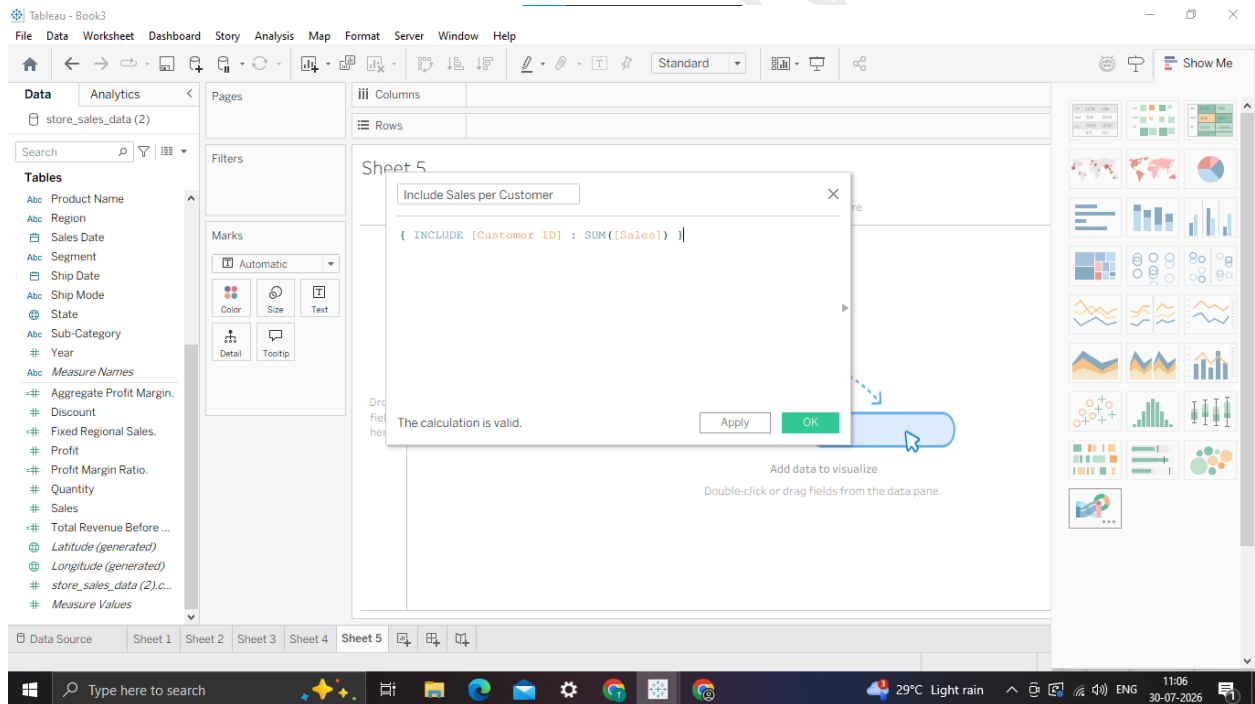
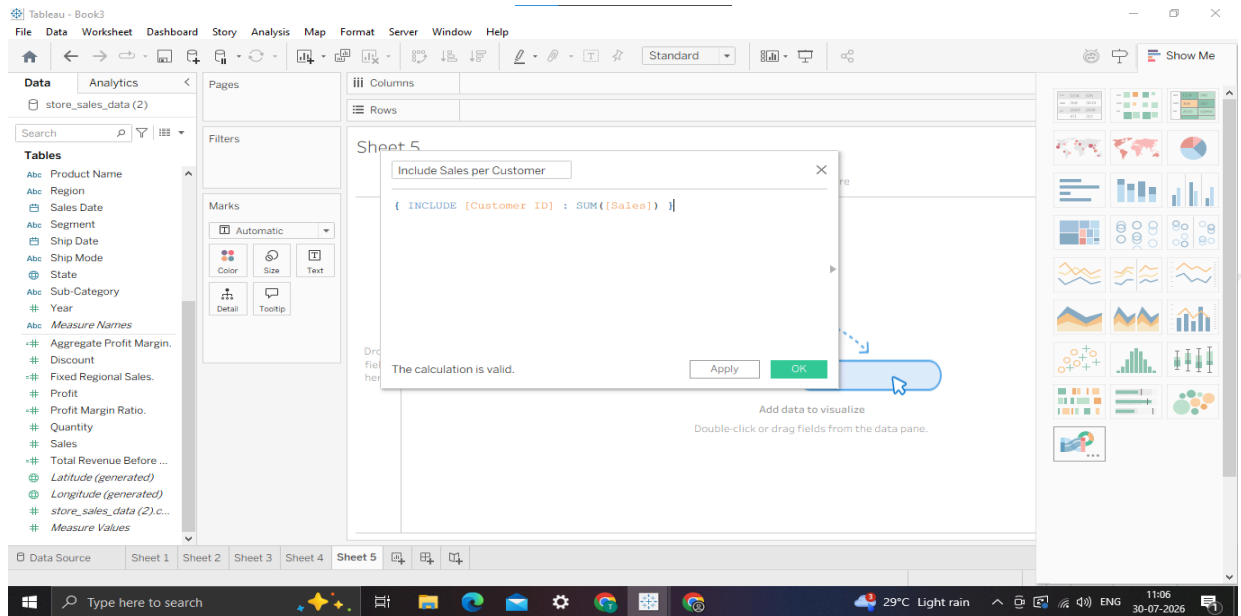


2. Putting the INCLUDE Field to Use

1. Drag the State dimension from the Data Pane and drop it onto the Rows shelf.
2. Drag your newly created Include Sales per Customer calculated field and drop it onto the Columns shelf.

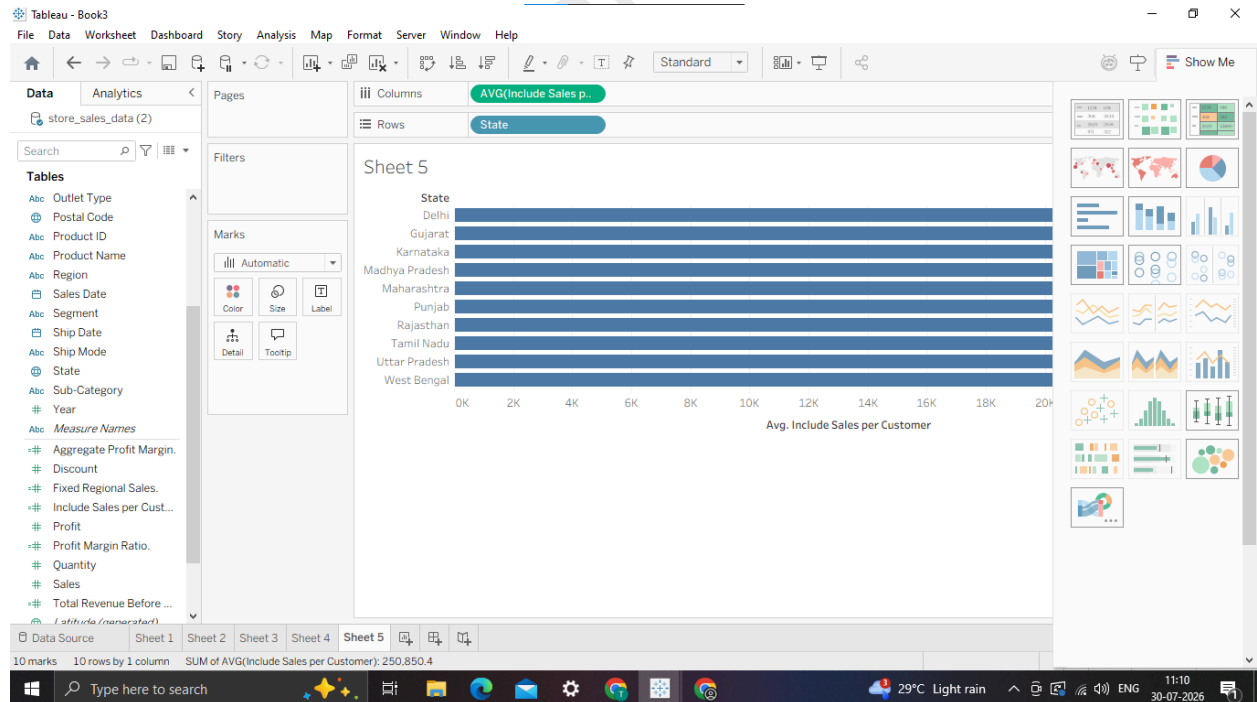
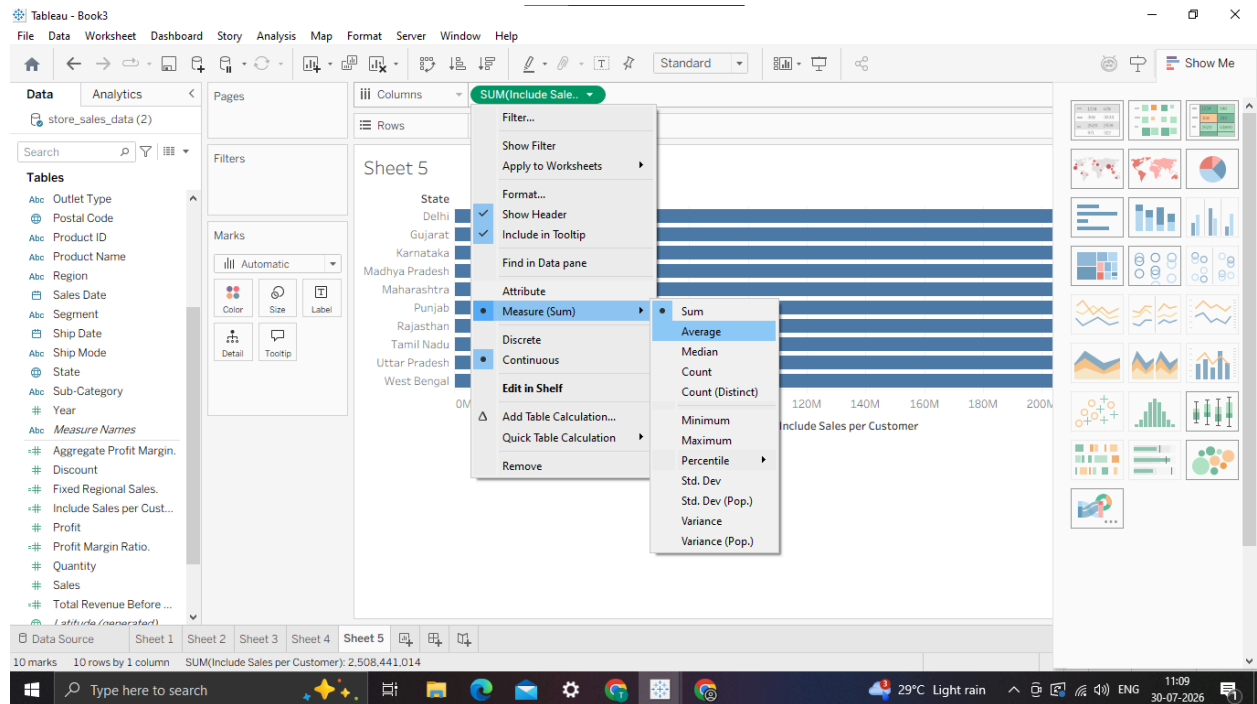


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3. Right-click the Include Sales per Customer pill on the Columns shelf, hover over Measure (Sum), and change it to Average.

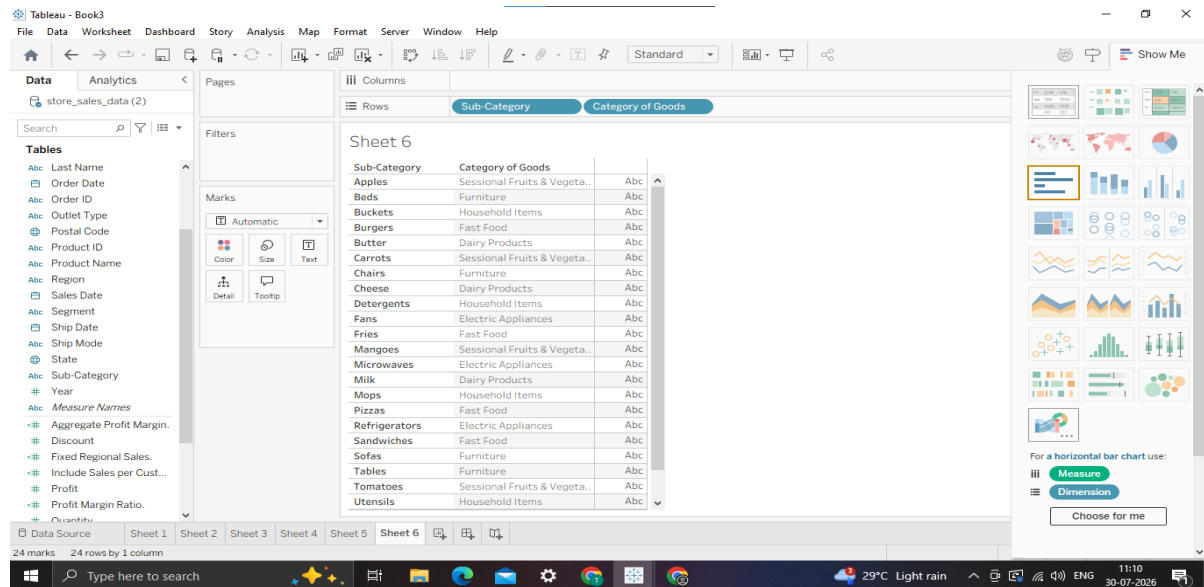


3. Creating an EXCLUDE LOD Calculated Field

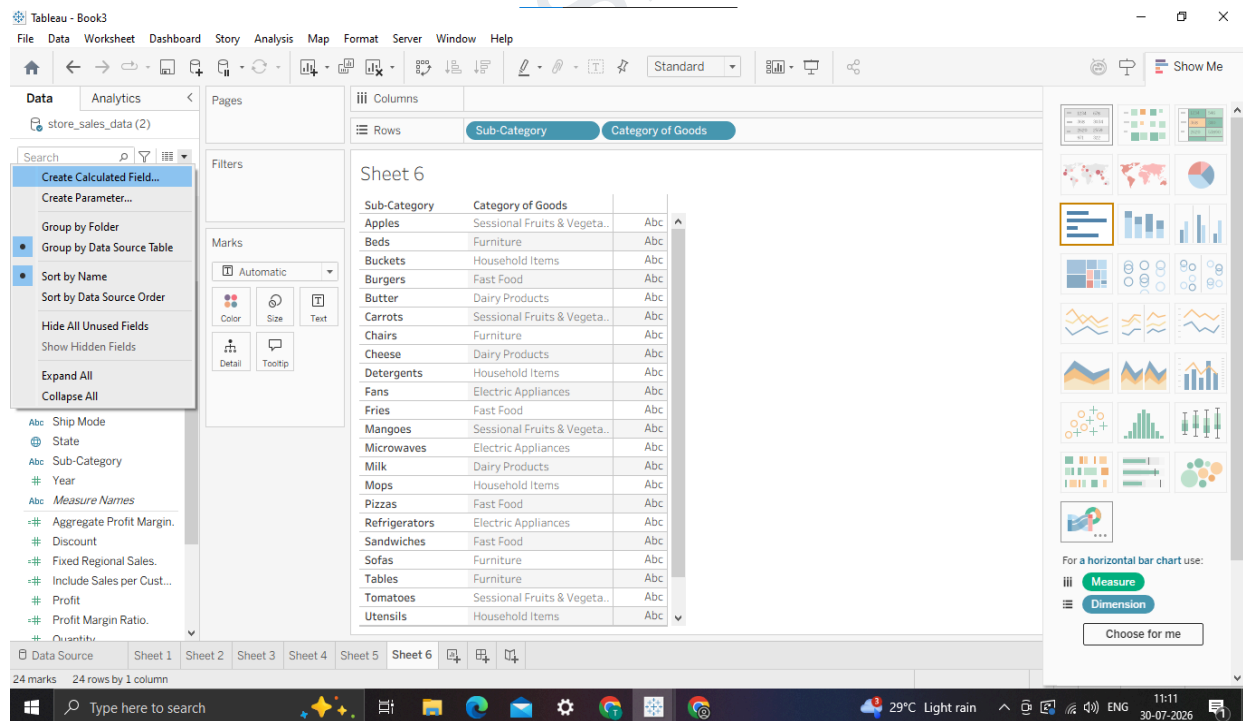
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1. Open a new blank worksheet.
2. Drag Category of Goods onto the Rows shelf.
3. Drag Sub-Category onto the Rows shelf (right next to Category of Goods).



4. Right-click in the blank white space of the Data Pane or Click the Small Triangle Icon next to search bar and select Create Calculated Field...



5. Name the field: Exclude Category Sales.

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Tableau - Book3

File Data Worksheet Dashboard Story Analysis Map Format Server Window Help

store_sales_data (2)

Search

Tables

- Abc Last Name
- Abc Order Date
- Abc Order ID
- Abc Outlet Type
- Abc Postal Code
- Abc Product ID
- Abc Product Name
- Abc Region
- Abc Sales Date
- Abc Segment
- Abc Ship Date
- Abc Ship Mode
- Abc State
- Abc Sub-Category
- # Year
- Abc Measure Names
- # Aggregate Profit Margin.
- # Discount
- # Fixed Regional Sales.
- # Include Sales per Cust...
- # Profit
- # Profit Margin Ratio.
- # Quantity

Filters

Marks

Automatic

Color Size Text

Detail Tooltip

Columns

Sub-Category Category of Goods

Sheet 6

Exclude Category Sales.

Apply OK

Sub-Category Category of Goods

Milk	Dairy Products	Abc
Mops	Household Items	Abc
Pizzas	Fast Food	Abc
Refrigerators	Electric Appliances	Abc
Sandwiches	Fast Food	Abc
Sofas	Furniture	Abc
Tables	Furniture	Abc
Tomatoes	Sessional Fruits & Vegeta...	Abc
Utensils	Household Items	Abc

24 marks 24 rows by 1 column

29°C Light rain 11:12 30-07-2026

6. Input the following expression:
`{ EXCLUDE [Sub-Category] : SUM([Sales]) }`

Tableau - Book3

File Data Worksheet Dashboard Story Analysis Map Format Server Window Help

store_sales_data (2)

Search

Tables

- Abc Last Name
- Abc Order Date
- Abc Order ID
- Abc Outlet Type
- Abc Postal Code
- Abc Product ID
- Abc Product Name
- Abc Region
- Abc Sales Date
- Abc Segment
- Abc Ship Date
- Abc Ship Mode
- Abc State
- Abc Sub-Category
- # Year
- Abc Measure Names
- # Aggregate Profit Margin.
- # Discount
- # Fixed Regional Sales.
- # Include Sales per Cust...
- # Profit
- # Profit Margin Ratio.
- # Quantity

Filters

Marks

Automatic

Color Size Text

Detail Tooltip

Columns

Sub-Category Category of Goods

Sheet 6

Exclude Category Sales.

{ EXCLUDE [Sub-Category] : SUM([Sales]) }

The calculation is valid.

Apply OK

Sub-Category Category of Goods

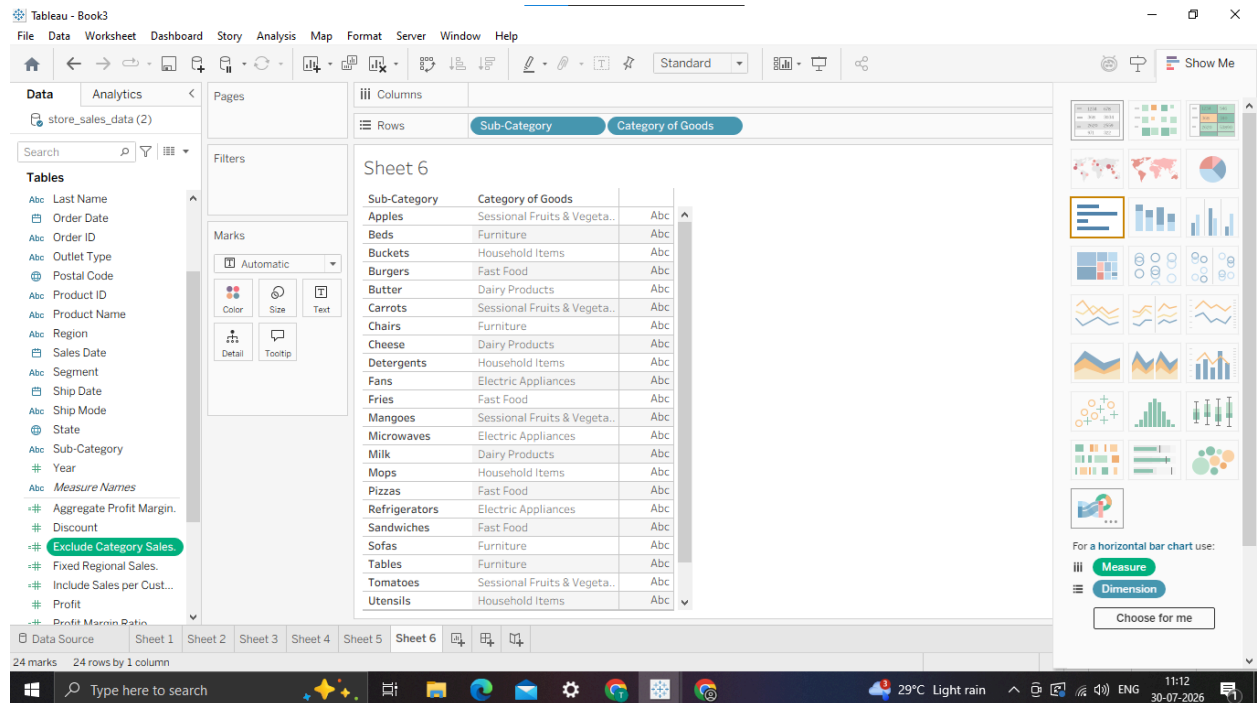
Milk	Dairy Products	Abc
Mops	Household Items	Abc
Pizzas	Fast Food	Abc
Refrigerators	Electric Appliances	Abc
Sandwiches	Fast Food	Abc
Sofas	Furniture	Abc
Tables	Furniture	Abc
Tomatoes	Sessional Fruits & Vegeta...	Abc
Utensils	Household Items	Abc

24 marks 24 rows by 1 column

29°C Light rain 11:12 30-07-2026

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7. Click OK and drag this new field onto the Columns shelf.



F: Creating and Using Parameters for Dynamic Analysis

1. Creating a Parameter

Unlike calculated fields which evaluate data statically, a Parameter is a dynamic variable or workbook input control block (like a slider or dropdown) that allows users to alter calculations or values on the fly without changing the underlying architecture.

Let's build a parameter that lets users set their own dynamic Profit Target threshold:

1. Go to your Data Pane. Click the small drop-down arrow (small triangle) next to the search bar at the top and choose Create Parameter.

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Tableau - Book3

File Data Worksheet Dashboard Story Analysis Map Format Server Window Help

Home

Navigation icons

Standard

Show Me

Data Analytics

store_sales_data (2)

Search

Create Calculated Field...

Create Parameter...

Group by Folder

Group by Data Source Table

Sort by Name

Sort by Data Source Order

Hide All Unused Fields

Show Hidden Fields

Expand All

Collapse All

Abc Ship Mode

Abc State

Abc Sub-Category

Abc Year

Abc Measure Names

Aggregate Profit Margin.

Discount

Exclude Category Sales.

Fixed Regional Sales.

Include Sales per Cust...

Profit

Profit Margin Ratio

Data Source

Sheet 1 Sheet 2 Sheet 3 Sheet 4 Sheet 5 Sheet 6

24 marks 24 rows by 1 column

Type here to search

29°C Light rain

ENG

11:13

30-07-2026

Sheet 6

Sub-Category	Category of Goods	
Apples	Sessional Fruits & Vegeta..	Abc
Beds	Furniture	Abc
Buckets	Household Items	Abc
Burgers	Fast Food	Abc
Butter	Dairy Products	Abc
Carrots	Sessional Fruits & Vegeta..	Abc
Chairs	Furniture	Abc
Cheese	Dairy Products	Abc
Detergents	Household Items	Abc
Fans	Electric Appliances	Abc
Fries	Fast Food	Abc
Mangoes	Sessional Fruits & Vegeta..	Abc
Microwaves	Electric Appliances	Abc
Milk	Dairy Products	Abc
Mops	Household Items	Abc
Pizzas	Fast Food	Abc
Refrigerators	Electric Appliances	Abc
Sandwiches	Fast Food	Abc
Sofas	Furniture	Abc
Tables	Furniture	Abc
Tomatoes	Sessional Fruits & Vegeta..	Abc
Utensils	Household Items	Abc

For a horizontal bar chart use:

Measure

Dimension

Choose for me

2. Name the parameter: Select Profit Threshold.

Tableau - Book3

File Data Worksheet Dashboard Story Analysis Map Format Server Window Help

Home

Navigation icons

Standard

Show Me

Data Analytics

store_sales_data (2)

Search

Tables

Abc Last Name

Abc Order Date

Abc Order ID

Abc Outlet Type

Abc Postal Code

Abc Product ID

Abc Product Name

Abc Region

Abc Sales Date

Abc Segment

Abc Ship Date

Abc Ship Mode

Abc State

Abc Sub-Category

Abc Year

Abc Measure Names

Aggregate Profit Margin.

Discount

Exclude Category Sales.

Fixed Regional Sales.

Include Sales per Cust...

Profit

Profit Margin Ratio

Data Source

Sheet 1 Sheet 2 Sheet 3 Sheet 4 Sheet 5 Sheet 6

24 marks 24 rows by 1 column

Type here to search

29°C Light rain

ENG

11:14

30-07-2026

Create Parameter

Name

Parameter 1

Properties

Data type

Float

Display format

1

Current value

1

Value when workbook opens

Current value

Allowable values

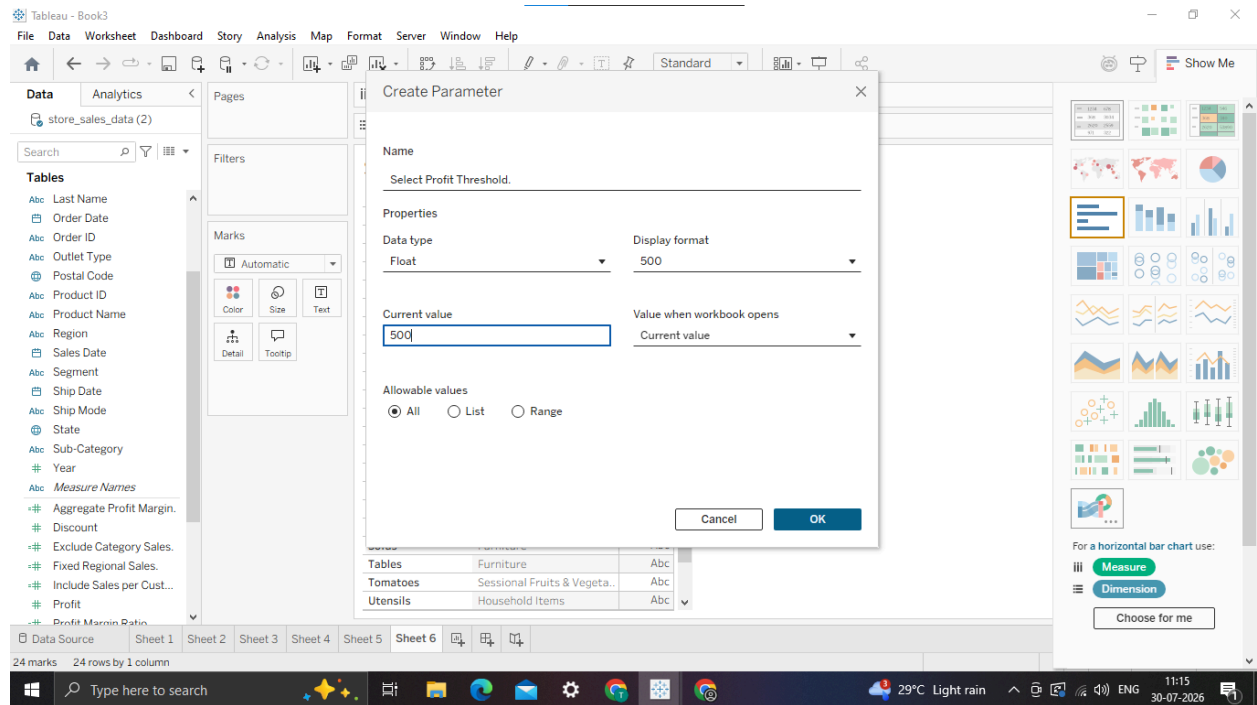
All List Range

Cancel OK

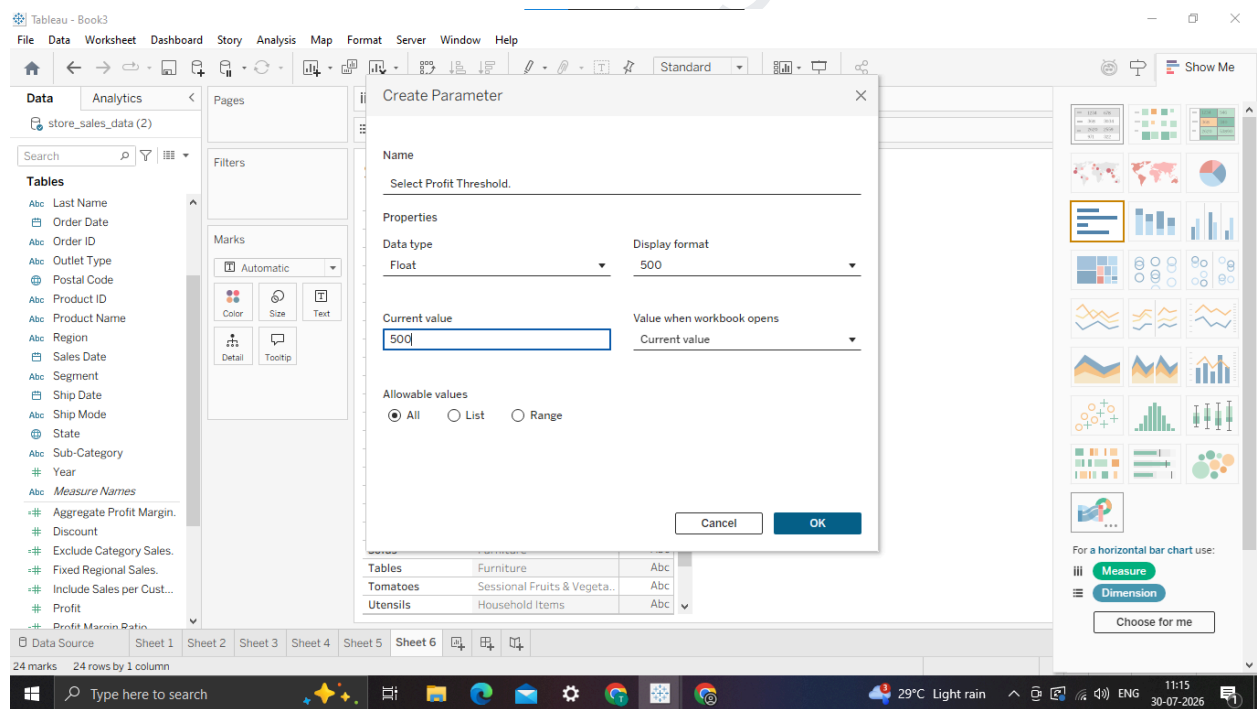
Tables

Tables		
Tables	Furniture	Abc
Tomatoes	Sessional Fruits & Vegeta..	Abc
Utensils	Household Items	Abc

3. Set the Data Type to Float (or Integer).

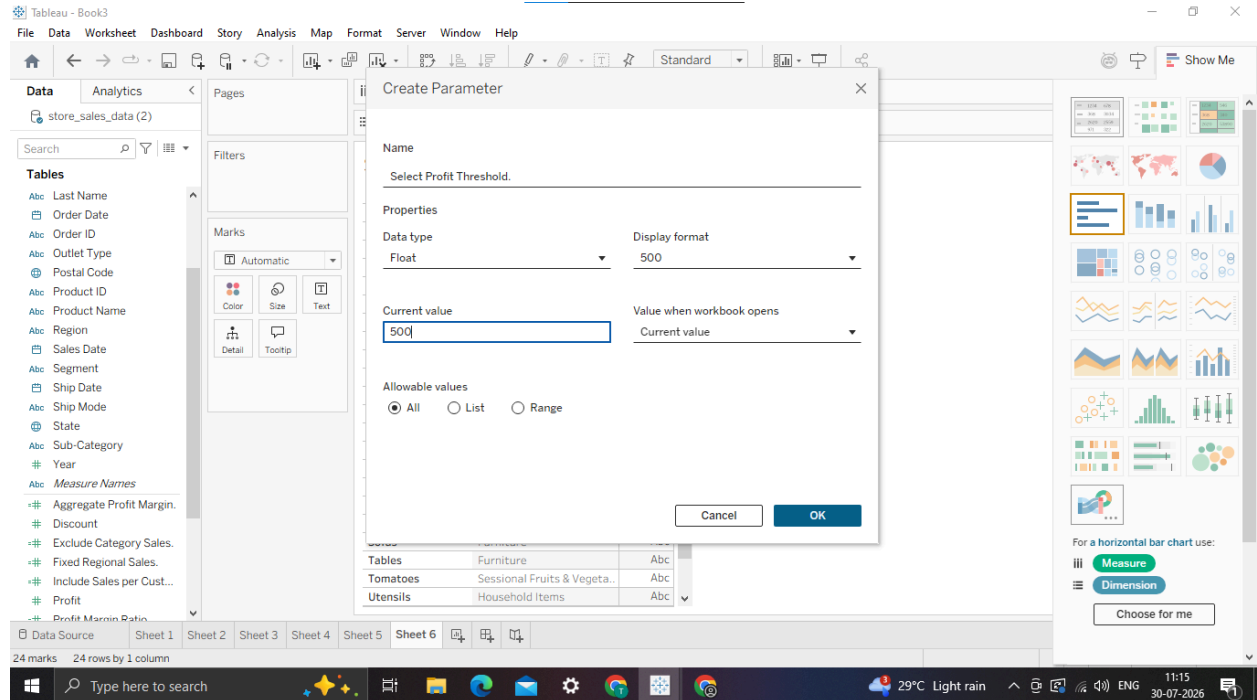


4. Set the Current Value to 500.

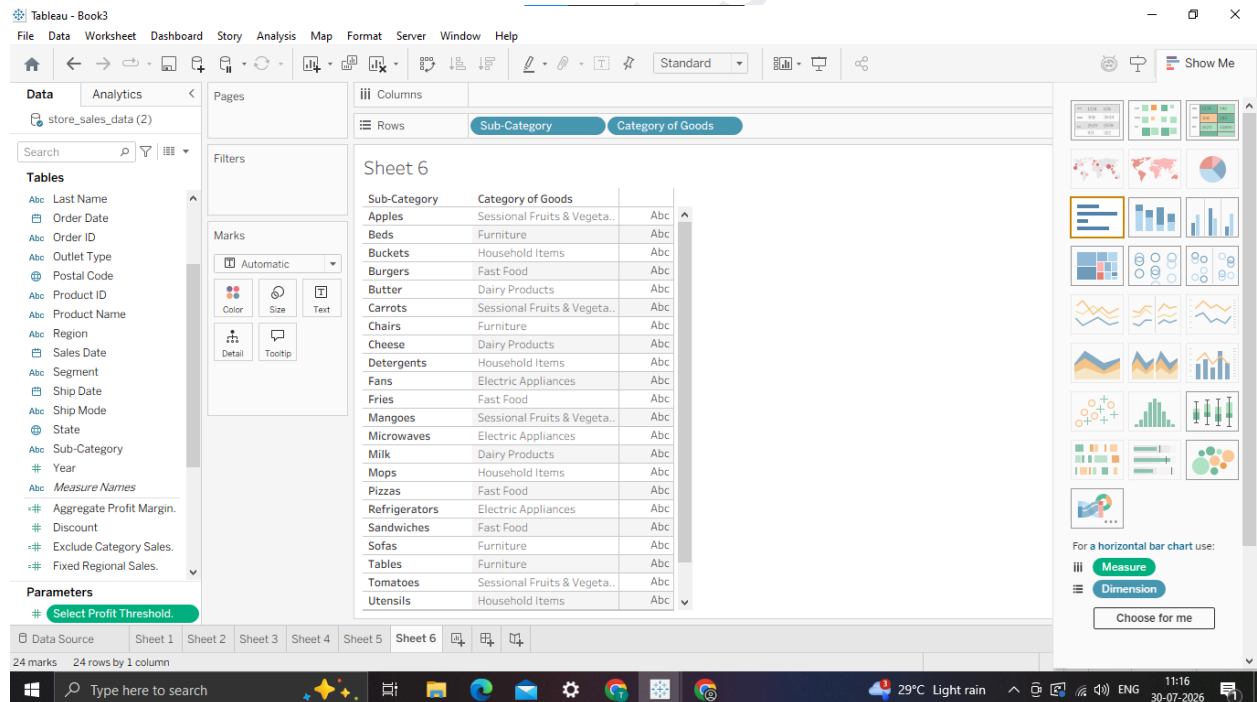


5. Keep the Allowable values set to All and click OK.

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6. Notice a new section titled Parameters appears at the bottom of your Data Pane, showcasing your new selector.



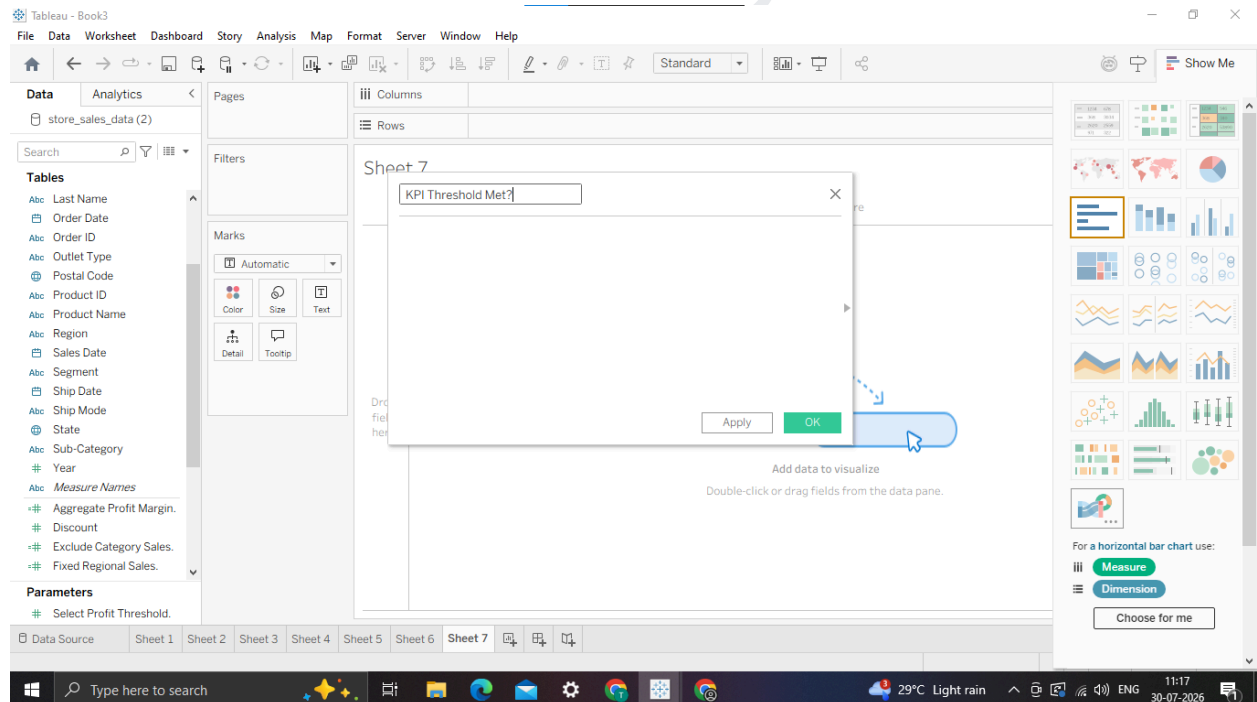
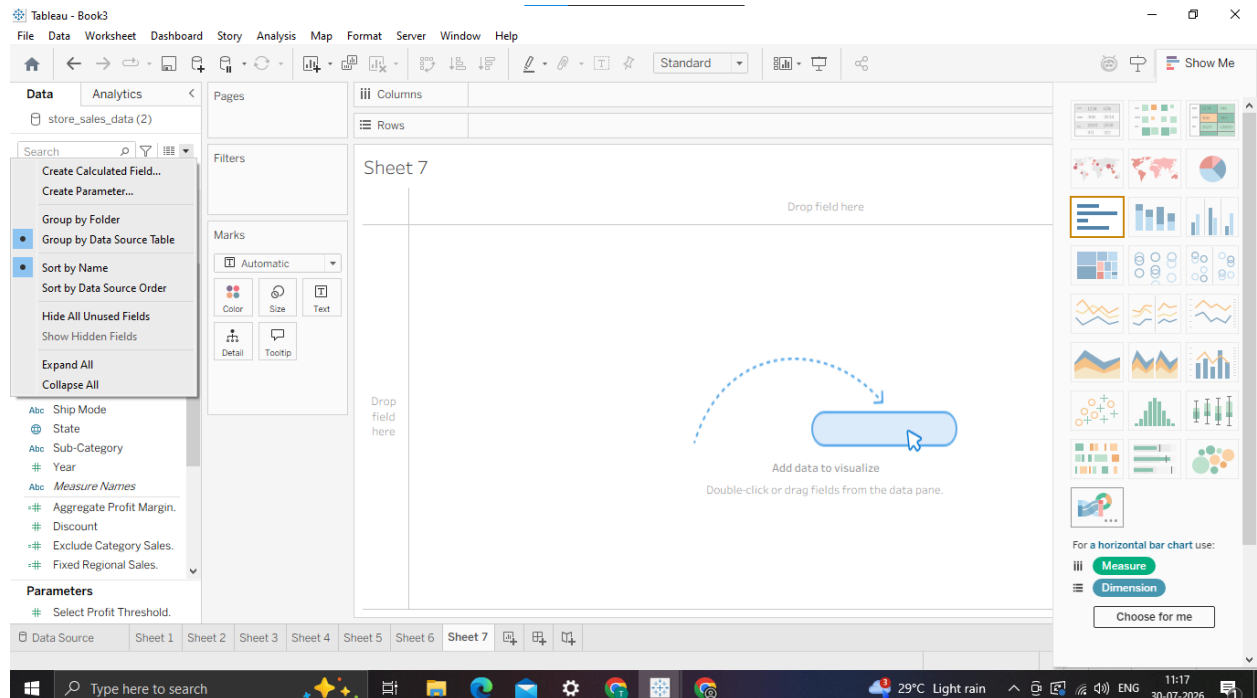
2. Linking the Parameter to a Calculated Field

A parameter does absolutely nothing until you actively tie it to a field layout or formula. Let's create a conditional formula that flags categories based on the parameter's current value:

1. Right-click in the blank white space of the Data Pane or Click on small triangle angle next to search bar and select Create Calculated Field...

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2. Name this field: KPI Threshold Met?.



3. Input the following logical IF-THEN conditional statement:

IF SUM([Profit]) >= [Select Profit Threshold] THEN "Target Achieved"

```
ELSE "Below Target"
```

END (NOTE: THE SELECT PROFIT THRESHOLD SHOULD TURN INTO PURPLE IF IT IS NOT TURNING IN PURPLE PLEASE RETYPE IT AGAIN